

# Hedge Overlay Recommendation

model\_v4 — "up-shock + vol-surge" 1-day semis hedge

*Standalone overlay (src/hedge\_overlay.py); the model is NOT modified.*

## THE RULE (decide at close T, hold T+1 only, exit next close)

FIRE if:   QQQ 1-day return  $\geq +2\%$    AND    $z(\text{SPY 5d realized vol, 63d}) \geq 1.0$   
THEN buy:   semis hedge sized INVERSE-VOL (hedge \$-vol = 50% of book \$-vol),  
              scaling with book size & relative vols -> avg notional:  
              short-SMH ~44%   or   SOXS ~14% (=same risk, -3x).

## WHAT IT TARGETS

- Sharp up-days that occur while volatility is already elevated -- a combination that tends to mean-revert (pull back) the next session.
- No single factor works alone (a QQQ up-day alone BOUNCES); the edge is the INTERACTION of up-shock x elevated vol.
- Best supporting signal (sensitivity): 5-day realized-vol z-score. Relative vol beats absolute level / surge-ratio; the 5d window beats 1d and 20d.

## QUANTIFIED VALUE (\$100k book, SOXS inverse-vol ~14% avg, 68 fires)

1-DAY hold (THE RULE):   net \$+26,846   [wins \$+47,962 / losses \$-21,116, 57% win]  
if held 5D:   \$+13,032       if held 20D:   \$-117,571   -> the 1-day exit IS the edge  
best saves:   2020-03-13 \$+7,369 (book -9.0%)   -   2026-06-16 \$+2,034 (book -3.8%)

## RISK-ADJUSTED (default short-SMH 0.30)

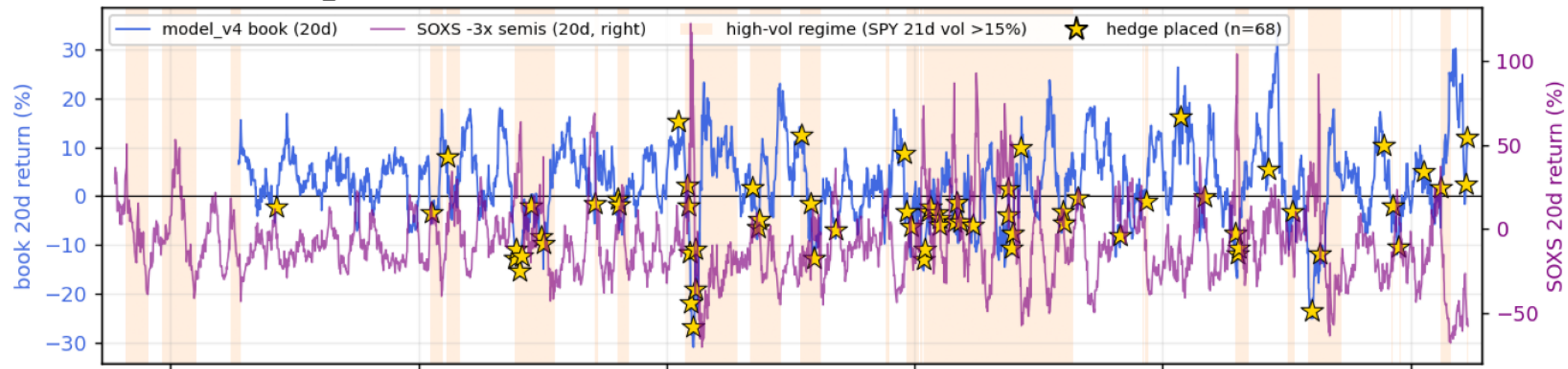
FULL:   dSharpe +0.11   sleeve PnL +24.7%   worst on-day -9.0% -> -4.7%  
OOS ( $\geq 2024$ ):   dSharpe +0.06   sleeve +3.5%  
Walk-forward: 7/10 active years dSharpe $>0$ ; gains concentrated in vol-spike years

## CAVEATS

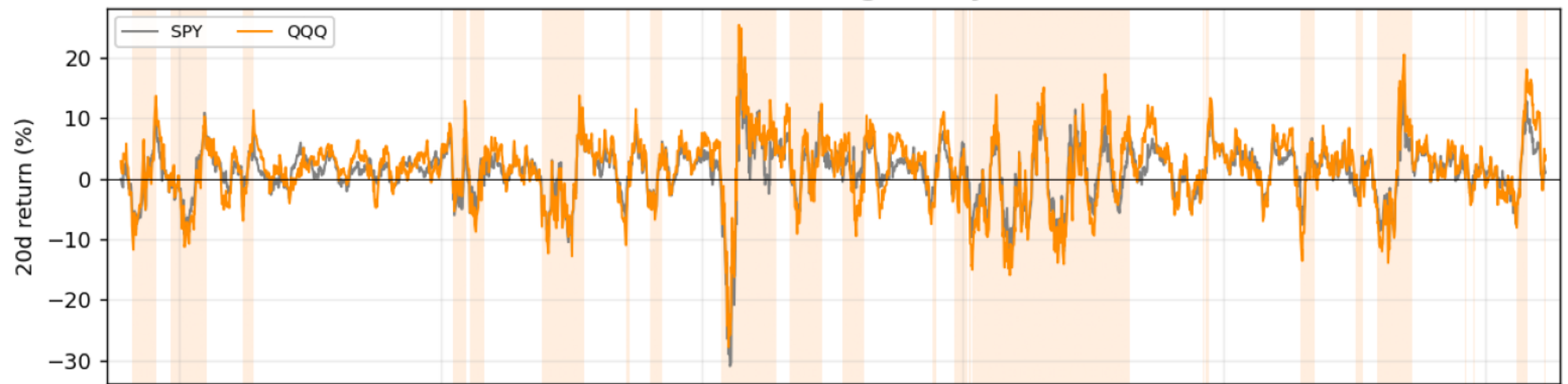
- A Sharpe-adder + single-day-blow softener, NOT an all-time-max-drawdown reducer.
- Edge concentrates in vol spikes; marginal in calm years; slightly -ve in 2022 bear.
- SOXS spread/decay understated by flat 10bp; short-SMH is cheaper & decay-free.
- Paper / recommendation only. Re-validate after any model or universe change.

SOXS hedge, inverse-vol sized (~14% avg notional) on \$100k book – PnL by hold horizon (n=68 fires)  
1-DAY (the rule): net \$+26,846 [wins \$+47,962 / losses \$-21,116, 57% win]  
5-DAY: \$+13,032 20-DAY: \$-117,571 -> strict 1-day exit is what saves \$  
best 1D saves: 2020-03-13 \$+7,369 (book -9.0%) - 2026-06-16 \$+2,034 (book -3.8%)

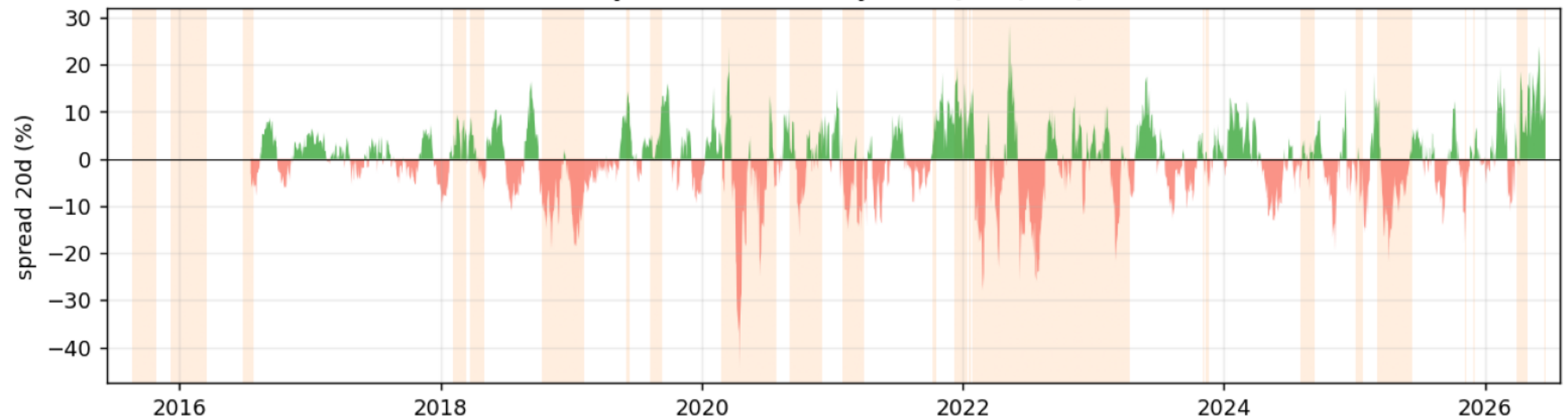
**model\_v4 portfolio vs SOXS hedge — ★ = hedge placed (up-shock + vol-surge)**



**SPY / QQQ — trailing 20-day return**



**Top10–Bottom10 spread (20d, fill)**



# Backtest — windows & yearly walk-forward

Default: short-SMH sized inverse-vol (hedge=50%% of book risk), trigger qqq\_1d>=2%% & z(spy\_vol\_5d)>=1. dSh = hedged-minus-book Sharpe; dDD = max-DD change (+ =better).

## WINDOW SUMMARY

| window      | days | on |  | bookSh | hedSh | dSh   |  | bookDD | dDD   |  | onBook% | paid% | sleeve% |
|-------------|------|----|--|--------|-------|-------|--|--------|-------|--|---------|-------|---------|
| FULL        | 2513 | 68 |  | +1.24  | +1.35 | +0.11 |  | -39.5% | -0.6% |  | -0.44   | 56    | +24.73% |
| COVID       | 1561 | 47 |  | +1.22  | +1.25 | +0.03 |  | -39.5% | -0.6% |  | 0.07    | 49    | +3.21%  |
| TRAIN(<oos) | 1896 | 52 |  | +1.18  | +1.30 | +0.13 |  | -38.1% | +1.4% |  | -0.58   | 60    | +21.21% |
| OOS         | 617  | 16 |  | +1.42  | +1.48 | +0.06 |  | -39.5% | -0.6% |  | -0.00   | 44    | +3.51%  |

## YEARLY WALK-FORWARD

| year | days | on |  | bookSh | hedSh | dSh   |  | bookDD | dDD    |  | onBook% | paid% | sleeve% |
|------|------|----|--|--------|-------|-------|--|--------|--------|--|---------|-------|---------|
| 2016 | 136  | 1  |  | +2.73  | +2.72 | -0.02 |  | -6.0%  | -0.0%  |  | -0.17   | 0     | -0.16%  |
| 2017 | 251  | 0  |  | +1.82  | +1.82 | +0.00 |  | -10.8% | +0.0%  |  | na      | na    | +0.00%  |
| 2018 | 251  | 8  |  | +0.71  | +0.80 | +0.09 |  | -28.9% | -0.1%  |  | -0.69   | 62    | +2.06%  |
| 2019 | 252  | 4  |  | +2.53  | +2.64 | +0.10 |  | -10.1% | +1.6%  |  | -0.80   | 75    | +1.44%  |
| 2020 | 253  | 11 |  | +1.44  | +1.99 | +0.55 |  | -33.8% | +10.6% |  | -2.03   | 73    | +17.10% |
| 2021 | 252  | 7  |  | +0.49  | +0.57 | +0.08 |  | -21.6% | +1.8%  |  | -0.41   | 71    | +2.17%  |
| 2022 | 251  | 16 |  | -0.76  | -0.82 | -0.05 |  | -31.8% | +0.4%  |  | 0.15    | 50    | -0.94%  |
| 2023 | 250  | 5  |  | +2.23  | +2.22 | -0.01 |  | -16.8% | -0.3%  |  | 0.33    | 40    | -0.46%  |
| 2024 | 252  | 6  |  | +2.12  | +2.16 | +0.04 |  | -20.4% | +1.9%  |  | 0.60    | 50    | +0.49%  |
| 2025 | 250  | 6  |  | -0.15  | -0.04 | +0.11 |  | -37.1% | -0.4%  |  | -0.44   | 50    | +3.45%  |
| 2026 | 115  | 4  |  | +3.01  | +3.02 | +0.01 |  | -12.1% | +0.0%  |  | -0.24   | 25    | -0.43%  |

### EXAMPLE CASE STUDY - 2026-06-16

Trigger @ 2026-06-15 close: QQQ +3.1% up-shock, SPY 5d-vol z = +1.46 (elevated) -> RULE FIRES  
Next day 2026-06-16: book -3.80% (leaders-led: top10 -5.7%, SMH -4.8%, QQQ -1.9%, SPY -0.6%; spread -4.0%)  
Hedge result: short-SMH (inv-vol, 42% wt) -> +1.99% (SOXS 12% wt -> +2.03%, \$+2,034 on \$100k) -> the -3.8% book day softened to ~ -1

# Appendix A

Correlation & Gap Analysis (full history 2016-2026)

*(source: corr\_tails\_by\_horizon.pdf)*

Trailing-factor avg, correlation & UP-DN gap, by forward book-move segment

UP/DN cells: top = mean of trailing measure in-segment; bottom = its corr with book fwd return (purple shade = |corr|).

D = UP avg - DN avg (pp for %-measures; raw for scores). D cells shaded amber by |D| within each column (deepest = biggest gap). See next page for sorted ranking.

|                     | FWD 1D           |                   |        | FWD 5D            |                   |        | FWD 20D          |                   |        |
|---------------------|------------------|-------------------|--------|-------------------|-------------------|--------|------------------|-------------------|--------|
|                     | thr +/-2.0%      |                   |        | thr +/-4.5%       |                   |        | thr +/-7.5%      |                   |        |
| TRAILING MEASURE    | UP n=296         | DN n=250          | D(U-D) | UP n=340          | DN n=222          | D(U-D) | UP n=655         | DN n=215          | D(U-D) |
| BOOK                |                  |                   |        |                   |                   |        |                  |                   |        |
| book_1d             | -0.2%<br>r=-0.05 | +0.0%<br>r=-0.03  | -0.3   | -0.1%<br>r=-0.01  | +0.2%<br>r=0.12   | -0.2   | +0.2%<br>r=-0.02 | +0.1%<br>r=0.02   | +0.1   |
| book_5d             | -0.1%<br>r=-0.03 | +0.2%<br>r=0.16   | -0.3   | -0.1%<br>r=-0.02  | +0.1%<br>r=0.02   | -0.2   | +0.9%<br>r=0.04  | +0.4%<br>r=-0.09  | +0.5   |
| book_20d            | +2.8%<br>r=-0.01 | +2.2%<br>r=0.14   | +0.6   | +2.8%<br>r=0.12   | +1.5%<br>r=0.06   | +1.3   | +3.0%<br>r=-0.03 | +1.2%<br>r=-0.08  | +1.8   |
| LONG/SHORT SPREAD   |                  |                   |        |                   |                   |        |                  |                   |        |
| spread_1d           | -0.2%<br>r=0.04  | -0.0%<br>r=-0.09  | -0.1   | -0.1%<br>r=-0.03  | +0.1%<br>r=-0.11  | -0.2   | -0.0%<br>r=-0.03 | +0.2%<br>r=-0.01  | -0.2   |
| spread_5d           | -0.1%<br>r=0.09  | +0.3%<br>r=-0.16  | -0.4   | -0.2%<br>r=-0.15  | -0.1%<br>r=-0.29  | -0.0   | -0.1%<br>r=-0.01 | -0.3%<br>r=-0.22  | +0.2   |
| spread_20d          | +0.2%<br>r=0.10  | +0.6%<br>r=-0.05  | -0.5   | +0.4%<br>r=-0.01  | +0.1%<br>r=-0.13  | +0.3   | -0.8%<br>r=0.11  | +0.1%<br>r=-0.08  | -0.9   |
| BASKETS             |                  |                   |        |                   |                   |        |                  |                   |        |
| top10_1d            | -0.3%<br>r=-0.06 | -0.0%<br>r=-0.03  | -0.3   | -0.0%<br>r=-0.02  | +0.1%<br>r=0.07   | -0.2   | +0.2%<br>r=-0.05 | +0.2%<br>r=0.01   | -0.0   |
| top10_5d            | -0.2%<br>r=0.03  | +0.2%<br>r=0.13   | -0.4   | -0.0%<br>r=-0.04  | +0.2%<br>r=-0.03  | -0.3   | +0.9%<br>r=0.01  | +0.6%<br>r=-0.13  | +0.3   |
| top10_20d           | +2.3%<br>r=0.01  | +2.2%<br>r=0.13   | +0.1   | +2.8%<br>r=0.09   | +1.7%<br>r=0.03   | +1.0   | +3.0%<br>r=-0.05 | +1.7%<br>r=-0.02  | +1.3   |
| bottom10_1d         | -0.1%<br>r=-0.09 | +0.0%<br>r=0.05   | -0.1   | +0.0%<br>r=0.01   | +0.0%<br>r=0.14   | +0.0   | +0.2%<br>r=-0.01 | +0.1%<br>r=0.01   | +0.2   |
| bottom10_5d         | -0.0%<br>r=-0.04 | -0.0%<br>r=0.24   | +0.0   | +0.1%<br>r=0.09   | +0.3%<br>r=0.22   | -0.2   | +1.0%<br>r=0.01  | +0.9%<br>r=0.06   | +0.1   |
| bottom10_20d        | +2.1%<br>r=-0.07 | +1.5%<br>r=0.17   | +0.6   | +2.3%<br>r=0.10   | +1.6%<br>r=0.15   | +0.8   | +3.8%<br>r=-0.14 | +1.6%<br>r=0.05   | +2.2   |
| universe_1d         | -0.2%<br>r=-0.08 | -0.0%<br>r=0.00   | -0.2   | -0.0%<br>r=0.03   | +0.1%<br>r=0.12   | -0.1   | +0.2%<br>r=-0.03 | +0.1%<br>r=0.04   | +0.1   |
| universe_5d         | -0.1%<br>r=-0.01 | +0.0%<br>r=0.21   | -0.1   | -0.1%<br>r=0.08   | +0.2%<br>r=0.16   | -0.3   | +0.8%<br>r=0.01  | +0.7%<br>r=0.03   | +0.1   |
| universe_20d        | +1.6%<br>r=-0.06 | +1.3%<br>r=0.18   | +0.3   | +1.9%<br>r=0.09   | +1.3%<br>r=0.13   | +0.6   | +2.8%<br>r=-0.14 | +1.8%<br>r=0.01   | +1.0   |
| ETFs                |                  |                   |        |                   |                   |        |                  |                   |        |
| spy_1d              | -0.2%<br>r=-0.11 | +0.0%<br>r=-0.03  | -0.3   | -0.1%<br>r=0.02   | +0.1%<br>r=0.10   | -0.2   | +0.1%<br>r=-0.04 | +0.0%<br>r=0.03   | +0.1   |
| spy_5d              | -0.2%<br>r=-0.08 | -0.1%<br>r=0.25   | -0.1   | -0.2%<br>r=0.03   | +0.0%<br>r=0.24   | -0.2   | +0.4%<br>r=0.02  | +0.2%<br>r=0.06   | +0.2   |
| spy_20d             | +0.3%<br>r=-0.13 | +0.2%<br>r=0.24   | +0.1   | +0.5%<br>r=0.09   | +0.3%<br>r=0.29   | +0.1   | +1.2%<br>r=-0.12 | +0.8%<br>r=0.10   | +0.5   |
| qqq_1d              | -0.3%<br>r=-0.11 | +0.0%<br>r=-0.02  | -0.3   | -0.1%<br>r=0.00   | +0.1%<br>r=0.09   | -0.2   | +0.1%<br>r=-0.04 | +0.1%<br>r=0.04   | +0.0   |
| qqq_5d              | -0.2%<br>r=-0.05 | +0.0%<br>r=0.20   | -0.2   | -0.2%<br>r=0.05   | +0.1%<br>r=0.18   | -0.3   | +0.4%<br>r=0.04  | +0.3%<br>r=0.00   | +0.1   |
| qqq_20d             | +0.6%<br>r=-0.09 | +0.6%<br>r=0.18   | +0.0   | +0.8%<br>r=0.12   | +0.8%<br>r=0.19   | -0.1   | +1.6%<br>r=-0.10 | +1.1%<br>r=0.02   | +0.5   |
| smh_1d              | -0.3%<br>r=-0.03 | +0.1%<br>r=-0.01  | -0.4   | +0.1%<br>r=-0.00  | +0.2%<br>r=0.07   | -0.2   | +0.2%<br>r=-0.02 | +0.1%<br>r=0.01   | +0.1   |
| smh_5d              | -0.0%<br>r=0.03  | +0.3%<br>r=0.18   | -0.3   | -0.0%<br>r=0.04   | +0.3%<br>r=0.17   | -0.3   | +0.7%<br>r=0.06  | +0.5%<br>r=-0.01  | +0.3   |
| smh_20d             | +2.2%<br>r=-0.01 | +1.8%<br>r=0.15   | +0.3   | +1.8%<br>r=0.15   | +1.4%<br>r=0.20   | +0.4   | +2.5%<br>r=-0.02 | +1.1%<br>r=0.20   | +1.3   |
| psq_1d              | +0.3%<br>r=0.10  | -0.0%<br>r=0.03   | +0.3   | +0.1%<br>r=-0.00  | -0.1%<br>r=-0.09  | +0.2   | -0.1%<br>r=0.04  | -0.0%<br>r=-0.04  | -0.0   |
| psq_5d              | +0.3%<br>r=0.05  | +0.0%<br>r=-0.18  | +0.3   | +0.2%<br>r=-0.04  | -0.1%<br>r=-0.16  | +0.4   | -0.4%<br>r=-0.04 | -0.2%<br>r=-0.01  | -0.1   |
| psq_20d             | -0.4%<br>r=0.09  | -0.4%<br>r=-0.19  | +0.0   | -0.6%<br>r=-0.13  | -0.6%<br>r=-0.20  | +0.1   | -1.4%<br>r=0.12  | -0.8%<br>r=-0.01  | -0.7   |
| soxs_1d             | +1.1%<br>r=0.03  | -0.2%<br>r=0.02   | +1.3   | -0.2%<br>r=0.01   | -0.7%<br>r=-0.05  | +0.6   | -0.6%<br>r=0.02  | -0.3%<br>r=-0.01  | -0.3   |
| soxs_5d             | +0.1%<br>r=-0.04 | -1.5%<br>r=-0.15  | +1.6   | -0.0%<br>r=-0.01  | -1.4%<br>r=-0.12  | +1.4   | -2.3%<br>r=-0.07 | -1.5%<br>r=-0.01  | -0.8   |
| soxs_20d            | -7.4%<br>r=0.02  | -6.8%<br>r=-0.16  | -0.5   | -6.1%<br>r=-0.12  | -5.9%<br>r=-0.20  | -0.2   | -8.3%<br>r=0.03  | -5.1%<br>r=-0.18  | -3.2   |
| AVG SCORE           |                  |                   |        |                   |                   |        |                  |                   |        |
| sc.top10_1d         | 0.835<br>r=0.08  | 0.835<br>r=0.01   | +0.001 | 0.836<br>r=-0.14  | 0.832<br>r=-0.12  | +0.004 | 0.836<br>r=0.15  | 0.831<br>r=-0.12  | +0.005 |
| sc.top10_5d         | 0.836<br>r=0.07  | 0.834<br>r=0.04   | +0.002 | 0.836<br>r=0.10   | 0.832<br>r=-0.03  | +0.004 | 0.835<br>r=0.13  | 0.830<br>r=-0.11  | +0.005 |
| sc.top10_20d        | 0.837<br>r=0.08  | 0.836<br>r=0.04   | +0.001 | 0.837<br>r=0.16   | 0.835<br>r=0.02   | +0.002 | 0.834<br>r=0.16  | 0.835<br>r=-0.14  | -0.001 |
| sc.bottom10_1d      | 0.209<br>r=-0.06 | 0.208<br>r=0.02   | +0.000 | 0.212<br>r=-0.11  | 0.212<br>r=0.20   | -0.001 | 0.210<br>r=-0.02 | 0.210<br>r=0.11   | -0.000 |
| sc.bottom10_5d      | 0.209<br>r=-0.05 | 0.207<br>r=-0.02  | +0.001 | 0.211<br>r=-0.05  | 0.210<br>r=0.14   | +0.001 | 0.210<br>r=-0.00 | 0.208<br>r=0.01   | +0.002 |
| sc.bottom10_20d     | 0.209<br>r=-0.01 | 0.208<br>r=0.00   | +0.000 | 0.209<br>r=0.05   | 0.208<br>r=-0.02  | +0.001 | 0.210<br>r=-0.06 | 0.207<br>r=0.08   | +0.003 |
| sc.universe_1d      | 0.507<br>r=-0.07 | 0.507<br>r=0.02   | -0.000 | 0.507<br>r=-0.09  | 0.506<br>r=-0.06  | +0.000 | 0.507<br>r=-0.31 | 0.506<br>r=-0.13  | +0.000 |
| sc.universe_5d      | 0.507<br>r=-0.07 | 0.507<br>r=0.01   | -0.000 | 0.507<br>r=-0.09  | 0.506<br>r=-0.06  | +0.000 | 0.507<br>r=-0.31 | 0.506<br>r=-0.13  | +0.000 |
| sc.universe_20d     | 0.506<br>r=-0.05 | 0.507<br>r=-0.00  | -0.000 | 0.507<br>r=-0.08  | 0.506<br>r=-0.06  | +0.000 | 0.507<br>r=-0.30 | 0.506<br>r=-0.12  | +0.000 |
| sc.held_1d          | 0.613<br>r=0.11  | 0.615<br>r=-0.06  | -0.001 | 0.612<br>r=0.03   | 0.610<br>r=-0.34  | +0.002 | 0.604<br>r=0.13  | 0.606<br>r=-0.28  | -0.002 |
| sc.held_5d          | 0.615<br>r=0.09  | 0.613<br>r=-0.07  | +0.001 | 0.613<br>r=0.06   | 0.609<br>r=-0.24  | +0.004 | 0.602<br>r=0.14  | 0.605<br>r=-0.21  | -0.002 |
| sc.held_20d         | 0.613<br>r=0.13  | 0.612<br>r=-0.05  | +0.001 | 0.608<br>r=0.18   | 0.608<br>r=-0.20  | +0.001 | 0.594<br>r=0.16  | 0.612<br>r=-0.05  | -0.018 |
| REALIZED VOL (ann.) |                  |                   |        |                   |                   |        |                  |                   |        |
| spy_vol_1d          | +14.1%<br>r=0.13 | +14.5%<br>r=-0.22 | -0.4   | +14.0%<br>r=0.00  | +15.3%<br>r=-0.32 | -1.3   | +13.4%<br>r=0.08 | +13.6%<br>r=-0.04 | -0.1   |
| spy_vol_5d          | +16.7%<br>r=0.20 | +17.7%<br>r=-0.26 | -1.0   | +16.9%<br>r=0.00  | +17.0%<br>r=-0.36 | -0.1   | +16.5%<br>r=0.12 | +15.1%<br>r=-0.05 | +1.4   |
| spy_vol_20d         | +17.8%<br>r=0.16 | +16.8%<br>r=-0.17 | +1.0   | +18.4%<br>r=0.02  | +15.5%<br>r=-0.30 | +2.9   | +17.5%<br>r=0.11 | +15.3%<br>r=-0.00 | +2.2   |
| qqq_vol_1d          | +18.3%<br>r=0.10 | +18.7%<br>r=-0.17 | -0.4   | +18.1%<br>r=-0.02 | +19.4%<br>r=-0.25 | -1.3   | +16.5%<br>r=0.09 | +18.5%<br>r=0.02  | -2.1   |
| qqq_vol_5d          | +21.6%<br>r=0.17 | +22.9%<br>r=-0.22 | -1.3   | +21.8%<br>r=-0.00 | +22.1%<br>r=-0.33 | -0.3   | +20.3%<br>r=0.14 | +21.1%<br>r=0.02  | -0.8   |
| qqq_vol_20d         | +22.8%<br>r=0.14 | +22.0%<br>r=-0.15 | +0.8   | +23.2%<br>r=0.02  | +21.0%<br>r=-0.27 | +2.2   | +21.6%<br>r=0.12 | +21.4%<br>r=0.06  | +0.2   |

Read across a horizon group: UP, DN, then gap D. Positive D = factor ran higher before up-moves. In-segment corr is range-compressed by tail conditioning.

# Biggest UP-DN trailing-factor gaps, sorted by |D| within each horizon

D = UP avg - DN avg (pp for %-measures, raw for scores). Each column sorted by absolute gap, largest first. Scores sit on a 0-1 scale, so their raw gaps look tiny next to % returns.

## FWD 1D (thr +/-2.0%)

| measure            | D    | (U avg / D avg) |
|--------------------|------|-----------------|
| 1.soxs_5d          | +1.6 | +0.1%/ -1.5%    |
| 2.soxs_1d          | +1.3 | +1.1%/ -0.2%    |
| 3.qqq_vol_5d       | -1.3 | +21.6%/+22.9%   |
| 4.spy_vol_5d       | -1.0 | +16.7%/+17.7%   |
| 5.spy_vol_20d      | +1.0 | +17.8%/+16.8%   |
| 6.qqq_vol_20d      | +0.8 | +22.8%/+22.0%   |
| 7.book_20d         | +0.6 | +2.8%/ +2.2%    |
| 8.bottom10_20d     | +0.6 | +2.1%/ +1.5%    |
| 9.soxs_20d         | -0.5 | -7.4%/ -6.8%    |
| 10.spread_20d      | -0.5 | +0.2%/ +0.6%    |
| 11.spy_vol_1d      | -0.4 | +14.1%/+14.5%   |
| 12.spread_5d       | -0.4 | -0.1%/ +0.3%    |
| 13.qqq_vol_1d      | -0.4 | +18.3%/+18.7%   |
| 14.top10_5d        | -0.4 | -0.2%/ +0.2%    |
| 15.smh_1d          | -0.4 | -0.3%/ +0.1%    |
| 16.smh_20d         | +0.3 | +2.2%/ +1.8%    |
| 17.book_5d         | -0.3 | -0.1%/ +0.2%    |
| 18.smh_5d          | -0.3 | -0.0%/ +0.3%    |
| 19.universe_20d    | +0.3 | +1.6%/ +1.3%    |
| 20.psq_5d          | +0.3 | +0.3%/ +0.0%    |
| 21.psq_1d          | +0.3 | +0.3%/ -0.0%    |
| 22.qqq_1d          | -0.3 | -0.3%/ +0.0%    |
| 23.top10_1d        | -0.3 | -0.3%/ -0.0%    |
| 24.spy_1d          | -0.3 | -0.2%/ +0.0%    |
| 25.book_1d         | -0.3 | -0.2%/ +0.0%    |
| 26.qqq_5d          | -0.2 | -0.2%/ +0.0%    |
| 27.universe_1d     | -0.2 | -0.2%/ -0.0%    |
| 28.sc.top10_5d     | +0.2 | +83.6%/+83.4%   |
| 29.sc.top10_20d    | +0.1 | +83.7%/+83.6%   |
| 30.spy_5d          | -0.1 | -0.2%/ -0.1%    |
| 31.sc.bottom10_5d  | +0.1 | +20.9%/+20.7%   |
| 32.bottom10_1d     | -0.1 | -0.1%/ +0.0%    |
| 33.spread_1d       | -0.1 | -0.2%/ -0.0%    |
| 34.sc.held_1d      | -0.1 | +61.3%/+61.5%   |
| 35.universe_5d     | -0.1 | -0.1%/ +0.0%    |
| 36.top10_20d       | +0.1 | +2.3%/ +2.2%    |
| 37.sc.held_5d      | +0.1 | +61.5%/+61.3%   |
| 38.sc.held_20d     | +0.1 | +61.3%/+61.2%   |
| 39.spy_20d         | +0.1 | +0.3%/ +0.2%    |
| 40.sc.top10_1d     | +0.1 | +83.5%/+83.5%   |
| 41.sc.bottom10_20d | +0.0 | +20.9%/+20.8%   |
| 42.sc.bottom10_1d  | +0.0 | +20.9%/+20.8%   |
| 43.psq_20d         | +0.0 | -0.4%/ -0.4%    |
| 44.bottom10_5d     | +0.0 | -0.0%/ -0.0%    |
| 45.sc.universe_20d | -0.0 | +50.6%/+50.7%   |
| 46.sc.universe_1d  | -0.0 | +50.7%/+50.7%   |
| 47.sc.universe_5d  | -0.0 | +50.7%/+50.7%   |
| 48.qqq_20d         | +0.0 | +0.6%/ +0.6%    |

## FWD 5D (thr +/-4.5%)

| measure            | D    | (U avg / D avg) |
|--------------------|------|-----------------|
| 1.spy_vol_20d      | +2.9 | +18.4%/+15.5%   |
| 2.qqq_vol_20d      | +2.2 | +23.2%/+21.0%   |
| 3.soxs_5d          | +1.4 | -0.0%/ -1.4%    |
| 4.book_20d         | +1.3 | +2.8%/ +1.5%    |
| 5.spy_vol_1d       | -1.3 | +14.0%/+15.3%   |
| 6.qqq_vol_1d       | -1.3 | +18.1%/+19.4%   |
| 7.top10_20d        | +1.0 | +2.8%/ +1.7%    |
| 8.bottom10_20d     | +0.8 | +2.3%/ +1.6%    |
| 9.universe_20d     | +0.6 | +1.9%/ +1.3%    |
| 10.soxs_1d         | +0.6 | -0.2%/ -0.7%    |
| 11.sc.held_5d      | +0.4 | +61.3%/+60.9%   |
| 12.sc.top10_1d     | +0.4 | +83.6%/+83.2%   |
| 13.smh_20d         | +0.4 | +1.8%/ +1.4%    |
| 14.sc.top10_5d     | +0.4 | +83.6%/+83.2%   |
| 15.psq_5d          | +0.4 | +0.2%/ -0.1%    |
| 16.qqq_5d          | -0.3 | -0.2%/ +0.1%    |
| 17.spread_20d      | +0.3 | +0.4%/ +0.1%    |
| 18.qqq_vol_5d      | -0.3 | +21.8%/+22.1%   |
| 19.universe_5d     | -0.3 | -0.1%/ +0.2%    |
| 20.smh_5d          | -0.3 | -0.0%/ +0.3%    |
| 21.top10_5d        | -0.3 | -0.0%/ +0.2%    |
| 22.book_1d         | -0.2 | -0.1%/ +0.2%    |
| 23.psq_1d          | +0.2 | +0.1%/ -0.1%    |
| 24.qqq_1d          | -0.2 | -0.1%/ +0.1%    |
| 25.bottom10_5d     | -0.2 | +0.1%/ +0.3%    |
| 26.book_5d         | -0.2 | -0.1%/ +0.1%    |
| 27.sc.top10_20d    | +0.2 | +83.7%/+83.5%   |
| 28.soxs_20d        | -0.2 | -6.1%/ -5.9%    |
| 29.spy_5d          | -0.2 | -0.2%/ +0.0%    |
| 30.spread_1d       | -0.2 | -0.1%/ +0.1%    |
| 31.smh_1d          | -0.2 | +0.1%/ +0.2%    |
| 32.spy_1d          | -0.2 | -0.1%/ +0.1%    |
| 33.top10_1d        | -0.2 | -0.0%/ +0.1%    |
| 34.sc.held_1d      | +0.2 | +61.2%/+61.0%   |
| 35.spy_20d         | +0.1 | +0.5%/ +0.3%    |
| 36.universe_1d     | -0.1 | -0.0%/ +0.1%    |
| 37.qqq_20d         | -0.1 | +0.8%/ +0.8%    |
| 38.sc.bottom10_20d | +0.1 | +20.9%/+20.8%   |
| 39.psq_20d         | +0.1 | -0.6%/ -0.6%    |
| 40.sc.bottom10_1d  | -0.1 | +21.2%/+21.2%   |
| 41.sc.held_20d     | +0.1 | +60.8%/+60.8%   |
| 42.sc.bottom10_5d  | +0.1 | +21.1%/+21.0%   |
| 43.spy_vol_5d      | -0.1 | +16.9%/+17.0%   |
| 44.spread_5d       | -0.0 | -0.2%/ -0.1%    |
| 45.bottom10_1d     | +0.0 | +0.0%/ +0.0%    |
| 46.sc.universe_5d  | +0.0 | +50.7%/+50.6%   |
| 47.sc.universe_1d  | +0.0 | +50.7%/+50.6%   |
| 48.sc.universe_20d | +0.0 | +50.7%/+50.6%   |

## FWD 20D (thr +/-7.5%)

| measure            | D    | (U avg / D avg) |
|--------------------|------|-----------------|
| 1.soxs_20d         | -3.2 | -8.3%/ -5.1%    |
| 2.bottom10_20d     | +2.2 | +3.8%/ +1.6%    |
| 3.spy_vol_20d      | +2.2 | +17.5%/+15.3%   |
| 4.qqq_vol_1d       | -2.1 | +16.5%/+18.5%   |
| 5.book_20d         | +1.8 | +3.0%/ +1.2%    |
| 6.sc.held_20d      | -1.8 | +59.4%/+61.2%   |
| 7.spy_vol_5d       | +1.4 | +16.5%/+15.1%   |
| 8.smh_20d          | +1.3 | +2.5%/ +1.1%    |
| 9.top10_20d        | +1.3 | +3.0%/ +1.7%    |
| 10.universe_20d    | +1.0 | +2.8%/ +1.8%    |
| 11.spread_20d      | -0.9 | -0.8%/ +0.1%    |
| 12.qqq_vol_5d      | -0.8 | +20.3%/+21.1%   |
| 13.soxs_5d         | -0.8 | -2.3%/ -1.5%    |
| 14.psq_20d         | -0.7 | -1.4%/ -0.8%    |
| 15.qqq_20d         | +0.5 | +1.6%/ +1.1%    |
| 16.book_5d         | +0.5 | +0.9%/ +0.4%    |
| 17.sc.top10_5d     | +0.5 | +83.5%/+83.0%   |
| 18.sc.top10_1d     | +0.5 | +83.6%/+83.1%   |
| 19.spy_20d         | +0.5 | +1.2%/ +0.8%    |
| 20.soxs_1d         | -0.3 | -0.6%/ -0.3%    |
| 21.sc.bottom10_20d | +0.3 | +21.0%/+20.7%   |
| 22.top10_5d        | +0.3 | +0.9%/ +0.6%    |
| 23.smh_5d          | +0.3 | +0.7%/ +0.5%    |
| 24.sc.held_1d      | -0.2 | +60.4%/+60.6%   |
| 25.sc.held_5d      | -0.2 | +60.2%/+60.5%   |
| 26.qqq_vol_20d     | +0.2 | +21.6%/+21.4%   |
| 27.spread_1d       | -0.2 | -0.0%/ +0.2%    |
| 28.spy_5d          | +0.2 | +0.4%/ +0.2%    |
| 29.sc.bottom10_5d  | +0.2 | +21.0%/+20.8%   |
| 30.bottom10_1d     | +0.2 | +0.2%/ +0.1%    |
| 31.spread_5d       | +0.2 | -0.1%/ -0.3%    |
| 32.spy_vol_1d      | -0.1 | +13.4%/+13.6%   |
| 33.psq_5d          | -0.1 | -0.4%/ -0.2%    |
| 34.bottom10_5d     | +0.1 | +1.0%/ +0.9%    |
| 35.sc.top10_20d    | -0.1 | +83.4%/+83.5%   |
| 36.smh_1d          | +0.1 | +0.2%/ +0.1%    |
| 37.qqq_5d          | +0.1 | +0.4%/ +0.3%    |
| 38.book_1d         | +0.1 | +0.2%/ +0.1%    |
| 39.universe_5d     | +0.1 | +0.8%/ +0.7%    |
| 40.spy_1d          | +0.1 | +0.1%/ +0.0%    |
| 41.universe_1d     | +0.1 | +0.2%/ +0.1%    |
| 42.psq_1d          | -0.0 | -0.1%/ -0.0%    |
| 43.qqq_1d          | +0.0 | +0.1%/ +0.1%    |
| 44.sc.universe_5d  | +0.0 | +50.7%/+50.6%   |
| 45.sc.universe_1d  | +0.0 | +50.7%/+50.6%   |
| 46.sc.universe_20d | +0.0 | +50.7%/+50.6%   |
| 47.top10_1d        | -0.0 | +0.2%/ +0.2%    |
| 48.sc.bottom10_1d  | -0.0 | +21.0%/+21.0%   |



Trailing -> Forward correlation call-outs (|r| >= 0.5)

Strong = |r| >= 0.70 | Moderate = 0.50 <= |r| < 0.70. Cells ranked by |r|.

avg() = mean of that factor across the days passing the panel filter.

Small-n panels (n < ~40) are noise-dominated; weight accordingly.

Forward 1-Day portfolio return (filtered by book\_fwd\_1d)

UP > +2.0% (n=296)

MODERATE (0.50-0.70):

|       |                  |            |    |                 |           |
|-------|------------------|------------|----|-----------------|-----------|
| -0.62 | spy_vol_trl_5d   | avg=+16.7% | -> | psq_fwd_1d      | avg=-1.7% |
| -0.61 | qqq_vol_trl_5d   | avg=+21.6% | -> | psq_fwd_1d      | avg=-1.7% |
| -0.58 | qqq_vol_trl_20d  | avg=+22.8% | -> | psq_fwd_1d      | avg=-1.7% |
| +0.58 | spy_trl_20d      | avg=+0.3%  | -> | psq_fwd_1d      | avg=-1.7% |
| +0.57 | qqq_vol_trl_20d  | avg=+22.8% | -> | universe_fwd_1d | avg=+2.3% |
| -0.57 | spy_vol_trl_20d  | avg=+17.8% | -> | psq_fwd_1d      | avg=-1.7% |
| +0.55 | qqq_vol_trl_5d   | avg=+21.6% | -> | universe_fwd_1d | avg=+2.3% |
| +0.55 | spy_vol_trl_5d   | avg=+16.7% | -> | universe_fwd_1d | avg=+2.3% |
| +0.54 | spy_vol_trl_20d  | avg=+17.8% | -> | universe_fwd_1d | avg=+2.3% |
| +0.54 | qqq_trl_20d      | avg=+0.6%  | -> | psq_fwd_1d      | avg=-1.7% |
| -0.54 | spy_trl_20d      | avg=+0.3%  | -> | universe_fwd_1d | avg=+2.3% |
| -0.54 | universe_trl_20d | avg=+1.6%  | -> | bottom10_fwd_1d | avg=+2.4% |
| +0.53 | universe_trl_20d | avg=+1.6%  | -> | psq_fwd_1d      | avg=-1.7% |
| -0.53 | psq_trl_20d      | avg=-0.4%  | -> | psq_fwd_1d      | avg=-1.7% |
| -0.53 | qqq_trl_20d      | avg=+0.6%  | -> | bottom10_fwd_1d | avg=+2.4% |
| -0.53 | book_trl_20d     | avg=+2.8%  | -> | bottom10_fwd_1d | avg=+2.4% |
| -0.53 | qqq_trl_20d      | avg=+0.6%  | -> | universe_fwd_1d | avg=+2.3% |
| -0.53 | universe_trl_20d | avg=+1.6%  | -> | universe_fwd_1d | avg=+2.3% |
| -0.52 | qqq_vol_trl_5d   | avg=+21.6% | -> | soxs_fwd_1d     | avg=-7.7% |
| +0.52 | psq_trl_20d      | avg=-0.4%  | -> | universe_fwd_1d | avg=+2.3% |
| +0.52 | book_trl_20d     | avg=+2.8%  | -> | psq_fwd_1d      | avg=-1.7% |
| +0.51 | psq_trl_20d      | avg=-0.4%  | -> | bottom10_fwd_1d | avg=+2.4% |
| +0.51 | qqq_vol_trl_20d  | avg=+22.8% | -> | bottom10_fwd_1d | avg=+2.4% |
| -0.51 | spy_vol_trl_5d   | avg=+16.7% | -> | soxs_fwd_1d     | avg=-7.7% |
| -0.50 | book_trl_20d     | avg=+2.8%  | -> | universe_fwd_1d | avg=+2.3% |
| +0.50 | qqq_vol_trl_5d   | avg=+21.6% | -> | smh_fwd_1d      | avg=+2.5% |

UP > +3.5% (n=68)

MODERATE (0.50-0.70):

|       |                  |            |    |                 |           |
|-------|------------------|------------|----|-----------------|-----------|
| -0.70 | book_trl_20d     | avg=+2.5%  | -> | bottom10_fwd_1d | avg=+3.1% |
| -0.67 | qqq_trl_20d      | avg=-0.3%  | -> | bottom10_fwd_1d | avg=+3.1% |
| -0.67 | qqq_vol_trl_20d  | avg=+25.1% | -> | psq_fwd_1d      | avg=-2.6% |
| +0.66 | book_trl_20d     | avg=+2.5%  | -> | spread_fwd_1d   | avg=+0.5% |
| -0.66 | universe_trl_20d | avg=+0.7%  | -> | bottom10_fwd_1d | avg=+3.1% |
| +0.66 | spy_trl_20d      | avg=-1.1%  | -> | psq_fwd_1d      | avg=-2.6% |
| -0.66 | spy_trl_20d      | avg=-1.1%  | -> | universe_fwd_1d | avg=+3.1% |
| -0.65 | qqq_trl_20d      | avg=-0.3%  | -> | universe_fwd_1d | avg=+3.1% |
| -0.65 | book_trl_20d     | avg=+2.5%  | -> | universe_fwd_1d | avg=+3.1% |
| -0.65 | universe_trl_20d | avg=+0.7%  | -> | universe_fwd_1d | avg=+3.1% |
| +0.65 | spy_trl_20d      | avg=-1.1%  | -> | spread_fwd_20d  | avg=-1.3% |
| -0.64 | spy_vol_trl_20d  | avg=+20.6% | -> | psq_fwd_1d      | avg=-2.6% |
| +0.64 | psq_trl_20d      | avg=+0.6%  | -> | bottom10_fwd_1d | avg=+3.1% |
| -0.64 | bottom10_trl_20d | avg=+0.7%  | -> | bottom10_fwd_1d | avg=+3.1% |
| +0.64 | psq_trl_20d      | avg=+0.6%  | -> | universe_fwd_1d | avg=+3.1% |
| -0.63 | spy_trl_20d      | avg=-1.1%  | -> | bottom10_fwd_1d | avg=+3.1% |
| -0.63 | bottom10_trl_20d | avg=+0.7%  | -> | universe_fwd_1d | avg=+3.1% |
| +0.63 | universe_trl_20d | avg=+0.7%  | -> | psq_fwd_1d      | avg=-2.6% |
| +0.63 | book_trl_20d     | avg=+2.5%  | -> | spread_fwd_20d  | avg=-1.3% |
| +0.63 | qqq_trl_20d      | avg=-0.3%  | -> | psq_fwd_1d      | avg=-2.6% |
| +0.63 | qqq_vol_trl_20d  | avg=+25.1% | -> | universe_fwd_1d | avg=+3.1% |
| -0.62 | spy_vol_trl_5d   | avg=+21.8% | -> | psq_fwd_1d      | avg=-2.6% |

|       |                     |            |    |                  |            |
|-------|---------------------|------------|----|------------------|------------|
| +0.62 | qqq_vol_trl_20d     | avg=+25.1% | -> | bottom10_fwd_1d  | avg=+3.1%  |
| +0.62 | book_trl_20d        | avg=+2.5%  | -> | psq_fwd_1d       | avg=-2.6%  |
| -0.61 | qqq_vol_trl_5d      | avg=+26.4% | -> | psq_fwd_1d       | avg=-2.6%  |
| +0.61 | spy_vol_trl_20d     | avg=+20.6% | -> | universe_fwd_1d  | avg=+3.1%  |
| -0.61 | psq_trl_20d         | avg=+0.6%  | -> | psq_fwd_1d       | avg=-2.6%  |
| +0.60 | qqq_trl_20d         | avg=-0.3%  | -> | spread_fwd_20d   | avg=-1.3%  |
| -0.60 | qqq_vol_trl_5d      | avg=+26.4% | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.60 | spy_vol_trl_20d     | avg=+20.6% | -> | bottom10_fwd_1d  | avg=+3.1%  |
| +0.60 | qqq_vol_trl_20d     | avg=+25.1% | -> | bottom10_fwd_20d | avg=+8.2%  |
| -0.60 | top10_trl_20d       | avg=+2.4%  | -> | bottom10_fwd_1d  | avg=+3.1%  |
| -0.60 | spy_vol_trl_5d      | avg=+21.8% | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.59 | spy_vol_trl_20d     | avg=+20.6% | -> | bottom10_fwd_20d | avg=+8.2%  |
| -0.58 | smh_trl_20d         | avg=+1.8%  | -> | bottom10_fwd_1d  | avg=+3.1%  |
| +0.58 | bottom10_trl_20d    | avg=+0.7%  | -> | psq_fwd_1d       | avg=-2.6%  |
| -0.58 | psq_trl_20d         | avg=+0.6%  | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.58 | qqq_trl_20d         | avg=-0.3%  | -> | spread_fwd_1d    | avg=+0.5%  |
| +0.57 | spy_trl_5d          | avg=-0.7%  | -> | psq_fwd_1d       | avg=-2.6%  |
| -0.57 | qqq_vol_trl_20d     | avg=+25.1% | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.57 | universe_trl_20d    | avg=+0.7%  | -> | spread_fwd_1d    | avg=+0.5%  |
| +0.57 | spy_vol_trl_5d      | avg=+21.8% | -> | universe_fwd_1d  | avg=+3.1%  |
| -0.57 | smh_trl_20d         | avg=+1.8%  | -> | universe_fwd_1d  | avg=+3.1%  |
| -0.57 | top10_trl_20d       | avg=+2.4%  | -> | universe_fwd_1d  | avg=+3.1%  |
| +0.56 | bottom10_trl_20d    | avg=+0.7%  | -> | spread_fwd_20d   | avg=-1.3%  |
| -0.56 | spy_vol_trl_20d     | avg=+20.6% | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.56 | qqq_vol_trl_5d      | avg=+26.4% | -> | universe_fwd_1d  | avg=+3.1%  |
| +0.56 | book_trl_5d         | avg=-0.1%  | -> | spread_fwd_1d    | avg=+0.5%  |
| +0.55 | smh_trl_20d         | avg=+1.8%  | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.55 | universe_trl_20d    | avg=+0.7%  | -> | spread_fwd_20d   | avg=-1.3%  |
| +0.55 | bottom10_trl_20d    | avg=+0.7%  | -> | spread_fwd_1d    | avg=+0.5%  |
| -0.54 | qqq_vol_trl_20d     | avg=+25.1% | -> | psq_fwd_20d      | avg=-4.7%  |
| +0.54 | qqq_vol_trl_20d     | avg=+25.1% | -> | smh_fwd_1d       | avg=+3.7%  |
| -0.54 | qqq_vol_trl_20d     | avg=+25.1% | -> | soxs_fwd_1d      | avg=-11.6% |
| -0.54 | spy_vol_trl_20d     | avg=+20.6% | -> | psq_fwd_20d      | avg=-4.7%  |
| -0.54 | spy_vol_trl_1d      | avg=+19.4% | -> | spread_fwd_20d   | avg=-1.3%  |
| -0.54 | psq_trl_20d         | avg=+0.6%  | -> | spread_fwd_1d    | avg=+0.5%  |
| +0.54 | book_trl_5d         | avg=-0.1%  | -> | psq_fwd_1d       | avg=-2.6%  |
| -0.53 | qqq_vol_trl_5d      | avg=+26.4% | -> | soxs_fwd_1d      | avg=-11.6% |
| +0.53 | smh_trl_20d         | avg=+1.8%  | -> | psq_fwd_1d       | avg=-2.6%  |
| +0.53 | top10_trl_20d       | avg=+2.4%  | -> | psq_fwd_1d       | avg=-2.6%  |
| +0.52 | spy_vol_trl_20d     | avg=+20.6% | -> | smh_fwd_1d       | avg=+3.7%  |
| -0.52 | spy_trl_20d         | avg=-1.1%  | -> | smh_fwd_1d       | avg=+3.7%  |
| +0.52 | top10_trl_20d       | avg=+2.4%  | -> | spread_fwd_1d    | avg=+0.5%  |
| -0.52 | spy_vol_trl_5d      | avg=+21.8% | -> | soxs_fwd_1d      | avg=-11.6% |
| +0.52 | spy_trl_20d         | avg=-1.1%  | -> | soxs_fwd_1d      | avg=-11.6% |
| +0.52 | smh_trl_20d         | avg=+1.8%  | -> | spread_fwd_1d    | avg=+0.5%  |
| -0.52 | spy_vol_trl_20d     | avg=+20.6% | -> | soxs_fwd_1d      | avg=-11.6% |
| -0.52 | spy_trl_5d          | avg=-0.7%  | -> | universe_fwd_1d  | avg=+3.1%  |
| -0.52 | score_top10_trl_20d | avg=0.840  | -> | universe_fwd_1d  | avg=+3.1%  |
| +0.52 | spy_vol_trl_5d      | avg=+21.8% | -> | bottom10_fwd_1d  | avg=+3.1%  |
| +0.52 | qqq_vol_trl_5d      | avg=+26.4% | -> | bottom10_fwd_20d | avg=+8.2%  |
| +0.52 | spy_vol_trl_5d      | avg=+21.8% | -> | bottom10_fwd_20d | avg=+8.2%  |
| +0.51 | qqq_trl_5d          | avg=-0.6%  | -> | psq_fwd_1d       | avg=-2.6%  |
| -0.51 | psq_trl_5d          | avg=+0.7%  | -> | psq_fwd_1d       | avg=-2.6%  |
| -0.51 | book_trl_5d         | avg=-0.1%  | -> | bottom10_fwd_1d  | avg=+3.1%  |
| +0.51 | qqq_vol_trl_5d      | avg=+26.4% | -> | bottom10_fwd_1d  | avg=+3.1%  |
| +0.50 | spy_trl_20d         | avg=-1.1%  | -> | spread_fwd_1d    | avg=+0.5%  |
| +0.50 | spy_vol_trl_5d      | avg=+21.8% | -> | smh_fwd_1d       | avg=+3.7%  |
| +0.50 | qqq_vol_trl_5d      | avg=+26.4% | -> | smh_fwd_1d       | avg=+3.7%  |



|       |                     |           |    |                 |           |
|-------|---------------------|-----------|----|-----------------|-----------|
| -0.50 | score_top10_trl_20d | avg=0.840 | -> | bottom10_fwd_1d | avg=+3.1% |
| -0.50 | universe_trl_20d    | avg=+0.7% | -> | smh_fwd_1d      | avg=+3.7% |

DOWN < -2.0% (n=250)

MODERATE (0.50-0.70):

|       |                |            |    |                 |           |
|-------|----------------|------------|----|-----------------|-----------|
| +0.53 | spy_vol_trl_5d | avg=+17.7% | -> | psq_fwd_1d      | avg=+2.1% |
| +0.51 | top10_trl_20d  | avg=+2.2%  | -> | bottom10_fwd_1d | avg=-2.7% |
| -0.51 | spy_vol_trl_5d | avg=+17.7% | -> | universe_fwd_1d | avg=-2.7% |
| +0.51 | qqq_vol_trl_5d | avg=+22.9% | -> | psq_fwd_1d      | avg=+2.1% |
| +0.51 | book_trl_20d   | avg=+2.2%  | -> | bottom10_fwd_1d | avg=-2.7% |

DOWN < -3.5% (n=77)

MODERATE (0.50-0.70):

|       |                 |            |    |                  |            |
|-------|-----------------|------------|----|------------------|------------|
| +0.63 | spy_vol_trl_5d  | avg=+21.9% | -> | psq_fwd_1d       | avg=+3.1%  |
| +0.62 | spy_vol_trl_1d  | avg=+17.4% | -> | psq_fwd_1d       | avg=+3.1%  |
| -0.62 | qqq_vol_trl_5d  | avg=+26.8% | -> | spread_fwd_20d   | avg=-2.9%  |
| +0.60 | qqq_vol_trl_5d  | avg=+26.8% | -> | psq_fwd_1d       | avg=+3.1%  |
| -0.59 | spy_vol_trl_5d  | avg=+21.9% | -> | spread_fwd_20d   | avg=-2.9%  |
| -0.58 | spy_vol_trl_1d  | avg=+17.4% | -> | universe_fwd_1d  | avg=-3.8%  |
| -0.58 | qqq_vol_trl_20d | avg=+24.0% | -> | spread_fwd_20d   | avg=-2.9%  |
| +0.57 | qqq_vol_trl_1d  | avg=+21.2% | -> | psq_fwd_1d       | avg=+3.1%  |
| -0.54 | spy_vol_trl_20d | avg=+19.1% | -> | spread_fwd_20d   | avg=-2.9%  |
| +0.53 | spy_vol_trl_20d | avg=+19.1% | -> | psq_fwd_1d       | avg=+3.1%  |
| -0.53 | spy_vol_trl_5d  | avg=+21.9% | -> | universe_fwd_1d  | avg=-3.8%  |
| -0.53 | qqq_vol_trl_1d  | avg=+21.2% | -> | universe_fwd_1d  | avg=-3.8%  |
| +0.52 | spy_vol_trl_5d  | avg=+21.9% | -> | bottom10_fwd_20d | avg=+1.4%  |
| +0.52 | spy_vol_trl_1d  | avg=+17.4% | -> | soxs_fwd_1d      | avg=+13.7% |
| +0.52 | qqq_vol_trl_20d | avg=+24.0% | -> | psq_fwd_1d       | avg=+3.1%  |
| +0.52 | spy_trl_20d     | avg=-1.0%  | -> | spread_fwd_20d   | avg=-2.9%  |
| -0.52 | spy_vol_trl_1d  | avg=+17.4% | -> | smh_fwd_1d       | avg=-4.4%  |
| +0.51 | spy_vol_trl_20d | avg=+19.1% | -> | bottom10_fwd_20d | avg=+1.4%  |
| -0.50 | smh_trl_20d     | avg=+0.9%  | -> | psq_fwd_1d       | avg=+3.1%  |
| +0.50 | book_trl_20d    | avg=+1.1%  | -> | bottom10_fwd_1d  | avg=-4.0%  |
| +0.50 | qqq_vol_trl_5d  | avg=+26.8% | -> | bottom10_fwd_20d | avg=+1.4%  |

Forward 5-Day portfolio return (filtered by book\_fwd\_5d)

UP > +4.5% (n=340)

MODERATE (0.50-0.70):

|       |                  |            |    |                  |           |
|-------|------------------|------------|----|------------------|-----------|
| -0.58 | qqq_vol_trl_20d  | avg=+23.2% | -> | psq_fwd_5d       | avg=-3.4% |
| +0.58 | qqq_vol_trl_20d  | avg=+23.2% | -> | bottom10_fwd_5d  | avg=+5.1% |
| +0.57 | qqq_vol_trl_20d  | avg=+23.2% | -> | universe_fwd_5d  | avg=+5.0% |
| -0.56 | spy_vol_trl_20d  | avg=+18.4% | -> | psq_fwd_5d       | avg=-3.4% |
| +0.56 | spy_vol_trl_20d  | avg=+18.4% | -> | bottom10_fwd_20d | avg=+7.9% |
| +0.55 | qqq_vol_trl_20d  | avg=+23.2% | -> | bottom10_fwd_20d | avg=+7.9% |
| +0.55 | spy_vol_trl_20d  | avg=+18.4% | -> | bottom10_fwd_5d  | avg=+5.1% |
| -0.54 | qqq_vol_trl_5d   | avg=+21.8% | -> | psq_fwd_5d       | avg=-3.4% |
| +0.54 | spy_vol_trl_5d   | avg=+16.9% | -> | bottom10_fwd_20d | avg=+7.9% |
| +0.54 | spy_vol_trl_20d  | avg=+18.4% | -> | universe_fwd_5d  | avg=+5.0% |
| +0.54 | book_trl_20d     | avg=+2.8%  | -> | spread_fwd_5d    | avg=+1.0% |
| -0.53 | spy_vol_trl_5d   | avg=+16.9% | -> | psq_fwd_5d       | avg=-3.4% |
| -0.53 | book_trl_20d     | avg=+2.8%  | -> | bottom10_fwd_5d  | avg=+5.1% |
| +0.52 | universe_trl_20d | avg=+1.9%  | -> | spread_fwd_5d    | avg=+1.0% |
| +0.52 | spy_vol_trl_5d   | avg=+16.9% | -> | universe_fwd_5d  | avg=+5.0% |
| +0.52 | qqq_vol_trl_5d   | avg=+21.8% | -> | universe_fwd_5d  | avg=+5.0% |
| -0.52 | universe_trl_20d | avg=+1.9%  | -> | bottom10_fwd_5d  | avg=+5.1% |
| +0.52 | qqq_vol_trl_5d   | avg=+21.8% | -> | bottom10_fwd_20d | avg=+7.9% |
| +0.52 | spy_vol_trl_5d   | avg=+16.9% | -> | bottom10_fwd_5d  | avg=+5.1% |
| +0.51 | qqq_vol_trl_5d   | avg=+21.8% | -> | bottom10_fwd_5d  | avg=+5.1% |
| +0.51 | qqq_trl_20d      | avg=+0.8%  | -> | spread_fwd_5d    | avg=+1.0% |

UP > +6.5% (n=127)

**MODERATE (0.50-0.70):**

|       |                  |            |    |                  |           |
|-------|------------------|------------|----|------------------|-----------|
| +0.66 | book_trl_20d     | avg=+3.9%  | -> | spread_fwd_5d    | avg=+2.0% |
| -0.65 | book_trl_20d     | avg=+3.9%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| -0.65 | qqq_vol_trl_5d   | avg=+21.9% | -> | psq_fwd_5d       | avg=-3.9% |
| +0.65 | spy_vol_trl_5d   | avg=+17.7% | -> | bottom10_fwd_20d | avg=+9.1% |
| +0.64 | qqq_vol_trl_20d  | avg=+25.3% | -> | bottom10_fwd_20d | avg=+9.1% |
| +0.64 | spy_vol_trl_20d  | avg=+20.7% | -> | bottom10_fwd_20d | avg=+9.1% |
| +0.63 | qqq_vol_trl_5d   | avg=+21.9% | -> | bottom10_fwd_20d | avg=+9.1% |
| -0.62 | qqq_vol_trl_20d  | avg=+25.3% | -> | psq_fwd_5d       | avg=-3.9% |
| -0.62 | spy_vol_trl_5d   | avg=+17.7% | -> | psq_fwd_5d       | avg=-3.9% |
| -0.61 | universe_trl_20d | avg=+2.8%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.60 | universe_trl_20d | avg=+2.8%  | -> | spread_fwd_5d    | avg=+2.0% |
| +0.60 | qqq_vol_trl_5d   | avg=+21.9% | -> | bottom10_fwd_5d  | avg=+5.9% |
| -0.60 | spy_vol_trl_20d  | avg=+20.7% | -> | psq_fwd_5d       | avg=-3.9% |
| +0.60 | qqq_vol_trl_20d  | avg=+25.3% | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.59 | spy_vol_trl_5d   | avg=+17.7% | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.59 | qqq_trl_20d      | avg=+1.6%  | -> | spread_fwd_5d    | avg=+2.0% |
| -0.58 | bottom10_trl_20d | avg=+3.2%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.58 | bottom10_trl_20d | avg=+3.2%  | -> | spread_fwd_5d    | avg=+2.0% |
| +0.57 | qqq_vol_trl_5d   | avg=+21.9% | -> | universe_fwd_5d  | avg=+5.8% |
| +0.57 | spy_trl_20d      | avg=+0.9%  | -> | spread_fwd_5d    | avg=+2.0% |
| +0.57 | spy_vol_trl_20d  | avg=+20.7% | -> | bottom10_fwd_5d  | avg=+5.9% |
| -0.57 | qqq_trl_20d      | avg=+1.6%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| -0.56 | spy_vol_trl_20d  | avg=+20.7% | -> | psq_fwd_20d      | avg=-5.1% |
| -0.56 | qqq_vol_trl_20d  | avg=+25.3% | -> | spread_fwd_5d    | avg=+2.0% |
| -0.56 | qqq_vol_trl_20d  | avg=+25.3% | -> | psq_fwd_20d      | avg=-5.1% |
| -0.56 | spy_trl_20d      | avg=+0.9%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.55 | spy_vol_trl_5d   | avg=+17.7% | -> | universe_fwd_5d  | avg=+5.8% |
| +0.55 | top10_trl_20d    | avg=+4.4%  | -> | spread_fwd_5d    | avg=+2.0% |
| -0.55 | spy_trl_20d      | avg=+0.9%  | -> | bottom10_fwd_20d | avg=+9.1% |
| +0.55 | spy_trl_20d      | avg=+0.9%  | -> | spread_fwd_20d   | avg=+1.8% |
| +0.55 | book_trl_20d     | avg=+3.9%  | -> | spread_fwd_20d   | avg=+1.8% |
| -0.55 | qqq_vol_trl_5d   | avg=+21.9% | -> | spread_fwd_5d    | avg=+2.0% |
| -0.55 | spy_vol_trl_5d   | avg=+17.7% | -> | psq_fwd_20d      | avg=-5.1% |
| -0.55 | book_trl_20d     | avg=+3.9%  | -> | universe_fwd_5d  | avg=+5.8% |
| +0.54 | book_trl_20d     | avg=+3.9%  | -> | psq_fwd_5d       | avg=-3.9% |
| -0.54 | spy_vol_trl_20d  | avg=+20.7% | -> | spread_fwd_5d    | avg=+2.0% |
| +0.54 | qqq_vol_trl_20d  | avg=+25.3% | -> | universe_fwd_5d  | avg=+5.8% |
| -0.54 | spy_vol_trl_5d   | avg=+17.7% | -> | spread_fwd_5d    | avg=+2.0% |
| -0.53 | book_trl_20d     | avg=+3.9%  | -> | bottom10_fwd_20d | avg=+9.1% |
| +0.53 | spy_trl_20d      | avg=+0.9%  | -> | psq_fwd_5d       | avg=-3.9% |
| -0.53 | psq_trl_20d      | avg=-1.5%  | -> | spread_fwd_5d    | avg=+2.0% |
| +0.53 | qqq_trl_20d      | avg=+1.6%  | -> | spread_fwd_20d   | avg=+1.8% |
| -0.53 | qqq_vol_trl_5d   | avg=+21.9% | -> | psq_fwd_20d      | avg=-5.1% |
| +0.53 | universe_trl_20d | avg=+2.8%  | -> | psq_fwd_5d       | avg=-3.9% |
| +0.52 | smh_trl_20d      | avg=+3.5%  | -> | spread_fwd_5d    | avg=+2.0% |
| -0.52 | universe_trl_20d | avg=+2.8%  | -> | universe_fwd_5d  | avg=+5.8% |
| +0.52 | smh_trl_20d      | avg=+3.5%  | -> | spread_fwd_20d   | avg=+1.8% |
| -0.52 | spy_trl_20d      | avg=+0.9%  | -> | universe_fwd_5d  | avg=+5.8% |
| -0.51 | top10_trl_20d    | avg=+4.4%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.51 | spy_vol_trl_20d  | avg=+20.7% | -> | universe_fwd_5d  | avg=+5.8% |
| +0.51 | psq_trl_20d      | avg=-1.5%  | -> | bottom10_fwd_5d  | avg=+5.9% |
| +0.50 | spy_vol_trl_5d   | avg=+17.7% | -> | universe_fwd_20d | avg=+8.3% |

**DOWN < -4.5% (n=222)**

**MODERATE (0.50-0.70):**

|       |                |            |    |                 |           |
|-------|----------------|------------|----|-----------------|-----------|
| -0.57 | qqq_vol_trl_5d | avg=+22.1% | -> | bottom10_fwd_5d | avg=-6.2% |
| -0.54 | qqq_vol_trl_5d | avg=+22.1% | -> | universe_fwd_5d | avg=-5.8% |
| -0.54 | spy_vol_trl_5d | avg=+17.0% | -> | bottom10_fwd_5d | avg=-6.2% |

|       |                 |            |    |                 |           |
|-------|-----------------|------------|----|-----------------|-----------|
| -0.53 | qqq_vol_trl_20d | avg=+21.0% | -> | bottom10_fwd_5d | avg=-6.2% |
| -0.53 | qqq_vol_trl_20d | avg=+21.0% | -> | universe_fwd_5d | avg=-5.8% |
| -0.52 | spy_vol_trl_5d  | avg=+17.0% | -> | universe_fwd_5d | avg=-5.8% |
| -0.51 | spy_vol_trl_20d | avg=+15.5% | -> | universe_fwd_5d | avg=-5.8% |
| -0.51 | spy_vol_trl_20d | avg=+15.5% | -> | bottom10_fwd_5d | avg=-6.2% |

DOWN < -6.5% (n=103)

MODERATE (0.50-0.70):

|       |                 |            |    |                 |           |
|-------|-----------------|------------|----|-----------------|-----------|
| -0.66 | qqq_vol_trl_20d | avg=+22.4% | -> | universe_fwd_5d | avg=-7.8% |
| -0.66 | qqq_vol_trl_20d | avg=+22.4% | -> | bottom10_fwd_5d | avg=-8.5% |
| -0.65 | spy_vol_trl_20d | avg=+16.8% | -> | universe_fwd_5d | avg=-7.8% |
| -0.65 | qqq_vol_trl_5d  | avg=+24.5% | -> | bottom10_fwd_5d | avg=-8.5% |
| -0.64 | spy_vol_trl_20d | avg=+16.8% | -> | bottom10_fwd_5d | avg=-8.5% |
| -0.62 | qqq_vol_trl_5d  | avg=+24.5% | -> | universe_fwd_5d | avg=-7.8% |
| -0.62 | spy_vol_trl_5d  | avg=+19.5% | -> | bottom10_fwd_5d | avg=-8.5% |
| -0.60 | qqq_vol_trl_20d | avg=+22.4% | -> | top10_fwd_5d    | avg=-8.1% |
| -0.60 | spy_vol_trl_5d  | avg=+19.5% | -> | universe_fwd_5d | avg=-7.8% |
| -0.56 | spy_vol_trl_20d | avg=+16.8% | -> | top10_fwd_5d    | avg=-8.1% |
| -0.56 | spy_vol_trl_1d  | avg=+18.2% | -> | bottom10_fwd_5d | avg=-8.5% |
| +0.55 | book_trl_5d     | avg=+0.2%  | -> | spread_fwd_20d  | avg=-3.6% |
| -0.55 | spy_vol_trl_20d | avg=+16.8% | -> | smh_fwd_5d      | avg=-8.1% |
| -0.54 | spy_vol_trl_1d  | avg=+18.2% | -> | universe_fwd_5d | avg=-7.8% |
| -0.53 | qqq_vol_trl_5d  | avg=+24.5% | -> | bottom10_fwd_1d | avg=-1.9% |
| +0.53 | spy_trl_5d      | avg=-0.3%  | -> | spread_fwd_20d  | avg=-3.6% |
| -0.53 | spy_vol_trl_5d  | avg=+19.5% | -> | bottom10_fwd_1d | avg=-1.9% |
| -0.53 | qqq_vol_trl_20d | avg=+22.4% | -> | smh_fwd_5d      | avg=-8.1% |
| -0.52 | qqq_vol_trl_1d  | avg=+22.2% | -> | bottom10_fwd_5d | avg=-8.5% |
| -0.52 | spy_vol_trl_5d  | avg=+19.5% | -> | smh_fwd_5d      | avg=-8.1% |
| -0.51 | qqq_vol_trl_5d  | avg=+24.5% | -> | smh_fwd_5d      | avg=-8.1% |

Forward 20-Day portfolio return (filtered by book\_fwd\_20d)

UP > +7.5% (n=655)

MODERATE (0.50-0.70):

|       |                 |            |    |                  |           |
|-------|-----------------|------------|----|------------------|-----------|
| +0.68 | qqq_vol_trl_20d | avg=+21.6% | -> | bottom10_fwd_20d | avg=+9.9% |
| +0.67 | spy_vol_trl_20d | avg=+17.5% | -> | bottom10_fwd_20d | avg=+9.9% |
| +0.65 | spy_vol_trl_5d  | avg=+16.5% | -> | bottom10_fwd_20d | avg=+9.9% |
| +0.64 | qqq_vol_trl_5d  | avg=+20.3% | -> | bottom10_fwd_20d | avg=+9.9% |
| -0.61 | spy_vol_trl_20d | avg=+17.5% | -> | psq_fwd_20d      | avg=-5.7% |
| -0.60 | spy_vol_trl_5d  | avg=+16.5% | -> | psq_fwd_20d      | avg=-5.7% |
| -0.59 | qqq_vol_trl_20d | avg=+21.6% | -> | psq_fwd_20d      | avg=-5.7% |
| +0.59 | qqq_vol_trl_20d | avg=+21.6% | -> | universe_fwd_20d | avg=+9.4% |
| +0.58 | spy_vol_trl_20d | avg=+17.5% | -> | universe_fwd_20d | avg=+9.4% |
| +0.57 | spy_vol_trl_5d  | avg=+16.5% | -> | universe_fwd_20d | avg=+9.4% |
| +0.57 | qqq_vol_trl_5d  | avg=+20.3% | -> | universe_fwd_20d | avg=+9.4% |
| -0.56 | qqq_vol_trl_5d  | avg=+20.3% | -> | psq_fwd_20d      | avg=-5.7% |

UP > +12.5% (n=268)

STRONG (|r|>=0.70):

|       |                 |            |    |                  |            |
|-------|-----------------|------------|----|------------------|------------|
| +0.76 | spy_vol_trl_5d  | avg=+19.7% | -> | bottom10_fwd_20d | avg=+12.2% |
| +0.75 | qqq_vol_trl_5d  | avg=+24.0% | -> | bottom10_fwd_20d | avg=+12.2% |
| +0.75 | qqq_vol_trl_20d | avg=+24.6% | -> | bottom10_fwd_20d | avg=+12.2% |
| +0.74 | spy_vol_trl_20d | avg=+20.2% | -> | bottom10_fwd_20d | avg=+12.2% |
| -0.73 | spy_trl_20d     | avg=+0.2%  | -> | bottom10_fwd_20d | avg=+12.2% |

MODERATE (0.50-0.70):

|       |                  |            |    |                  |            |
|-------|------------------|------------|----|------------------|------------|
| -0.66 | spy_vol_trl_20d  | avg=+20.2% | -> | psq_fwd_20d      | avg=-7.0%  |
| +0.66 | spy_trl_20d      | avg=+0.2%  | -> | psq_fwd_20d      | avg=-7.0%  |
| -0.66 | universe_trl_20d | avg=+1.2%  | -> | bottom10_fwd_20d | avg=+12.2% |
| -0.65 | qqq_vol_trl_20d  | avg=+24.6% | -> | psq_fwd_20d      | avg=-7.0%  |
| -0.65 | spy_vol_trl_5d   | avg=+19.7% | -> | psq_fwd_20d      | avg=-7.0%  |
| +0.65 | qqq_vol_trl_5d   | avg=+24.0% | -> | universe_fwd_20d | avg=+11.9% |

|       |                  |            |    |                  |            |
|-------|------------------|------------|----|------------------|------------|
| -0.65 | bottom10_trl_20d | avg=+1.7%  | -> | bottom10_fwd_20d | avg=+12.2% |
| +0.65 | spy_vol_trl_5d   | avg=+19.7% | -> | universe_fwd_20d | avg=+11.9% |
| +0.63 | universe_trl_20d | avg=+1.2%  | -> | psq_fwd_20d      | avg=-7.0%  |
| -0.62 | book_trl_20d     | avg=+1.8%  | -> | bottom10_fwd_20d | avg=+12.2% |
| +0.62 | spy_trl_20d      | avg=+0.2%  | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.62 | qqq_vol_trl_20d  | avg=+24.6% | -> | universe_fwd_20d | avg=+11.9% |
| -0.62 | qqq_vol_trl_5d   | avg=+24.0% | -> | psq_fwd_20d      | avg=-7.0%  |
| +0.62 | spy_vol_trl_20d  | avg=+20.2% | -> | universe_fwd_20d | avg=+11.9% |
| -0.61 | qqq_trl_20d      | avg=+0.6%  | -> | bottom10_fwd_20d | avg=+12.2% |
| -0.60 | spy_vol_trl_5d   | avg=+19.7% | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.59 | book_trl_20d     | avg=+1.8%  | -> | spread_fwd_20d   | avg=+3.6%  |
| -0.59 | spy_trl_20d      | avg=+0.2%  | -> | universe_fwd_20d | avg=+11.9% |
| +0.59 | qqq_trl_20d      | avg=+0.6%  | -> | psq_fwd_20d      | avg=-7.0%  |
| -0.59 | qqq_vol_trl_20d  | avg=+24.6% | -> | spread_fwd_20d   | avg=+3.6%  |
| -0.59 | qqq_vol_trl_5d   | avg=+24.0% | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.58 | psq_trl_20d      | avg=-0.4%  | -> | bottom10_fwd_20d | avg=+12.2% |
| -0.58 | spy_vol_trl_20d  | avg=+20.2% | -> | spread_fwd_20d   | avg=+3.6%  |
| -0.57 | universe_trl_20d | avg=+1.2%  | -> | universe_fwd_20d | avg=+11.9% |
| +0.57 | top10_trl_20d    | avg=+1.5%  | -> | spread_fwd_20d   | avg=+3.6%  |
| -0.56 | psq_trl_20d      | avg=-0.4%  | -> | psq_fwd_20d      | avg=-7.0%  |
| +0.55 | bottom10_trl_20d | avg=+1.7%  | -> | psq_fwd_20d      | avg=-7.0%  |
| +0.55 | qqq_trl_20d      | avg=+0.6%  | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.54 | spy_vol_trl_1d   | avg=+15.2% | -> | bottom10_fwd_20d | avg=+12.2% |
| -0.53 | qqq_trl_20d      | avg=+0.6%  | -> | universe_fwd_20d | avg=+11.9% |
| +0.53 | book_trl_20d     | avg=+1.8%  | -> | psq_fwd_20d      | avg=-7.0%  |
| -0.52 | bottom10_trl_20d | avg=+1.7%  | -> | universe_fwd_20d | avg=+11.9% |
| +0.52 | spy_trl_20d      | avg=+0.2%  | -> | spread_fwd_5d    | avg=+0.4%  |
| -0.52 | spy_trl_20d      | avg=+0.2%  | -> | bottom10_fwd_5d  | avg=+3.6%  |
| -0.51 | psq_trl_20d      | avg=-0.4%  | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.51 | smh_trl_20d      | avg=+1.4%  | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.51 | universe_trl_20d | avg=+1.2%  | -> | spread_fwd_20d   | avg=+3.6%  |
| +0.50 | psq_trl_20d      | avg=-0.4%  | -> | universe_fwd_20d | avg=+11.9% |
| -0.50 | book_trl_20d     | avg=+1.8%  | -> | universe_fwd_20d | avg=+11.9% |

DOWN < -7.5% (n=215)

(none |r|>=0.5)

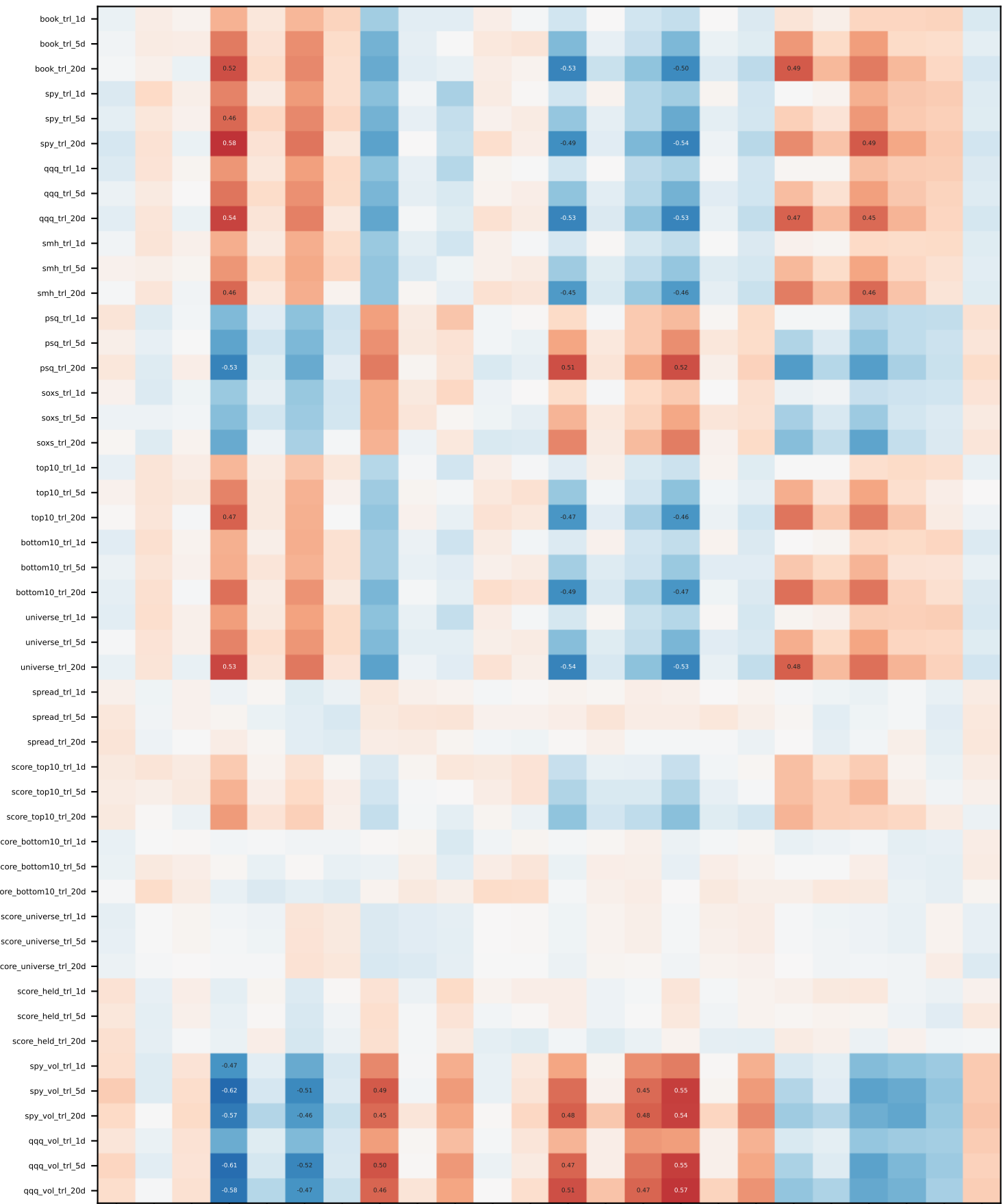
DOWN < -12.5% (n=70)

MODERATE (0.50-0.70):

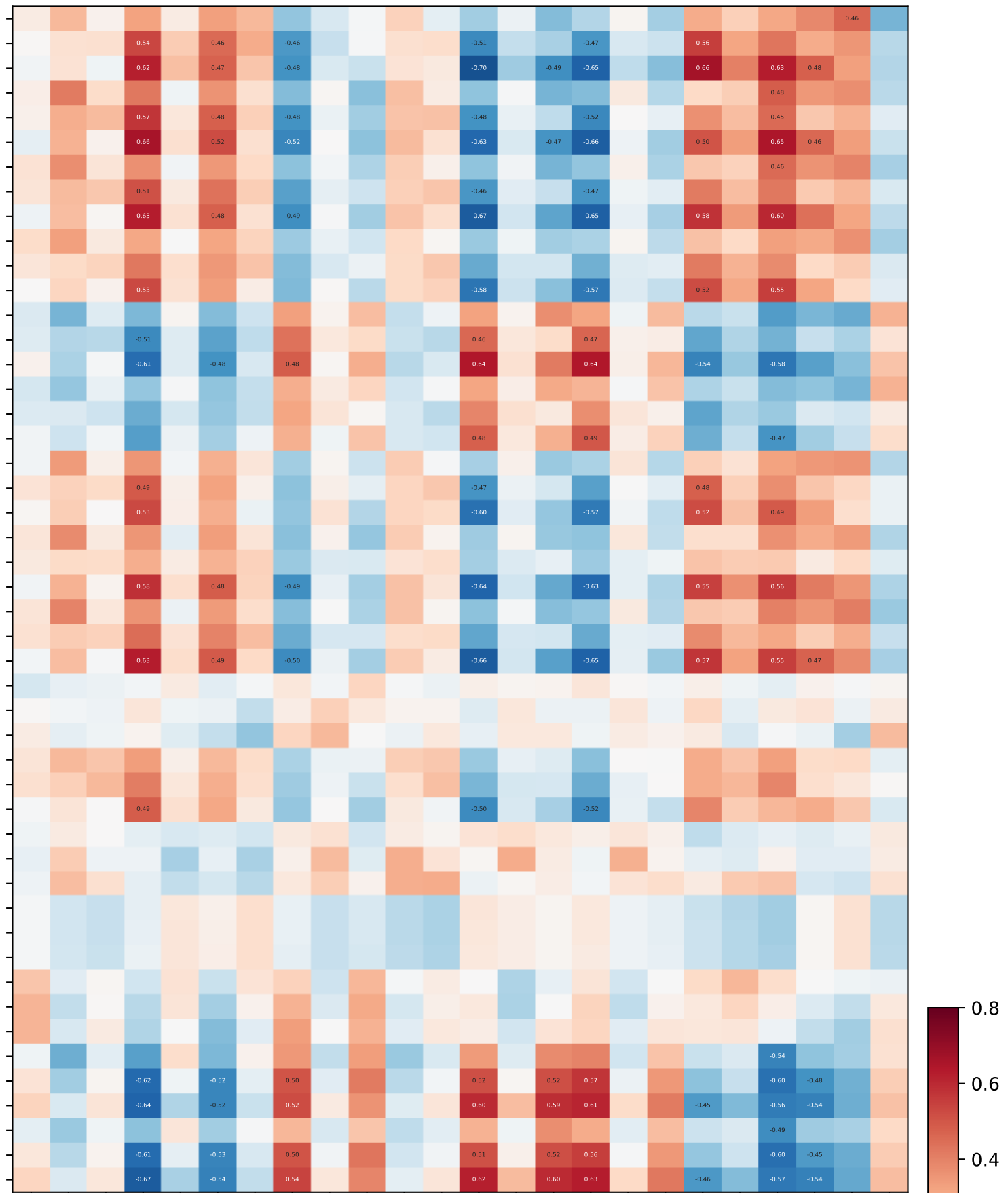
|       |                    |            |    |                 |            |
|-------|--------------------|------------|----|-----------------|------------|
| -0.58 | psq_trl_20d        | avg=+0.1%  | -> | spread_fwd_20d  | avg=-1.2%  |
| -0.57 | spy_vol_trl_5d     | avg=+15.7% | -> | bottom10_fwd_5d | avg=-3.6%  |
| +0.57 | book_trl_20d       | avg=-0.1%  | -> | spread_fwd_20d  | avg=-1.2%  |
| -0.57 | spy_vol_trl_5d     | avg=+15.7% | -> | soxs_fwd_20d    | avg=+35.9% |
| -0.56 | qqq_vol_trl_5d     | avg=+20.3% | -> | soxs_fwd_20d    | avg=+35.9% |
| +0.55 | spy_trl_20d        | avg=-0.2%  | -> | soxs_fwd_20d    | avg=+35.9% |
| +0.55 | qqq_trl_20d        | avg=+0.3%  | -> | spread_fwd_20d  | avg=-1.2%  |
| -0.54 | spy_vol_trl_20d    | avg=+14.8% | -> | bottom10_fwd_5d | avg=-3.6%  |
| -0.53 | qqq_vol_trl_20d    | avg=+20.2% | -> | soxs_fwd_20d    | avg=+35.9% |
| -0.52 | spy_vol_trl_20d    | avg=+14.8% | -> | soxs_fwd_20d    | avg=+35.9% |
| -0.52 | qqq_vol_trl_5d     | avg=+20.3% | -> | bottom10_fwd_5d | avg=-3.6%  |
| +0.52 | book_trl_5d        | avg=+0.2%  | -> | spread_fwd_20d  | avg=-1.2%  |
| -0.51 | score_top10_trl_1d | avg=0.831  | -> | book_fwd_20d    | avg=-17.7% |
| -0.51 | soxs_trl_20d       | avg=+1.5%  | -> | spread_fwd_20d  | avg=-1.2%  |
| +0.50 | universe_trl_20d   | avg=+0.6%  | -> | spread_fwd_20d  | avg=-1.2%  |

Trailing (rows) vs Forward (cols) correlations — filtered by Forward 1-Day portfolio return | scale +/-0.8

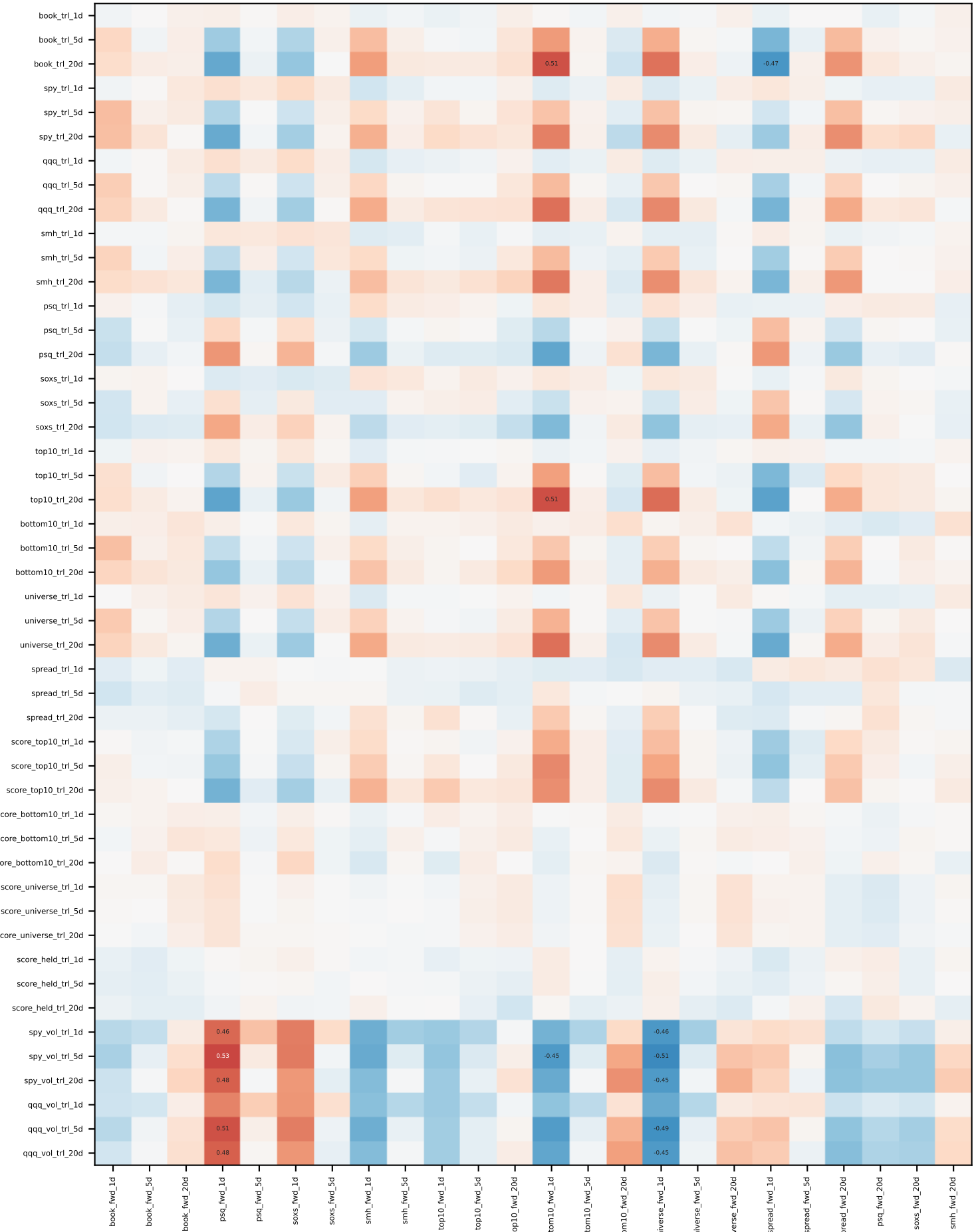
UP > +2.0% (n=296)



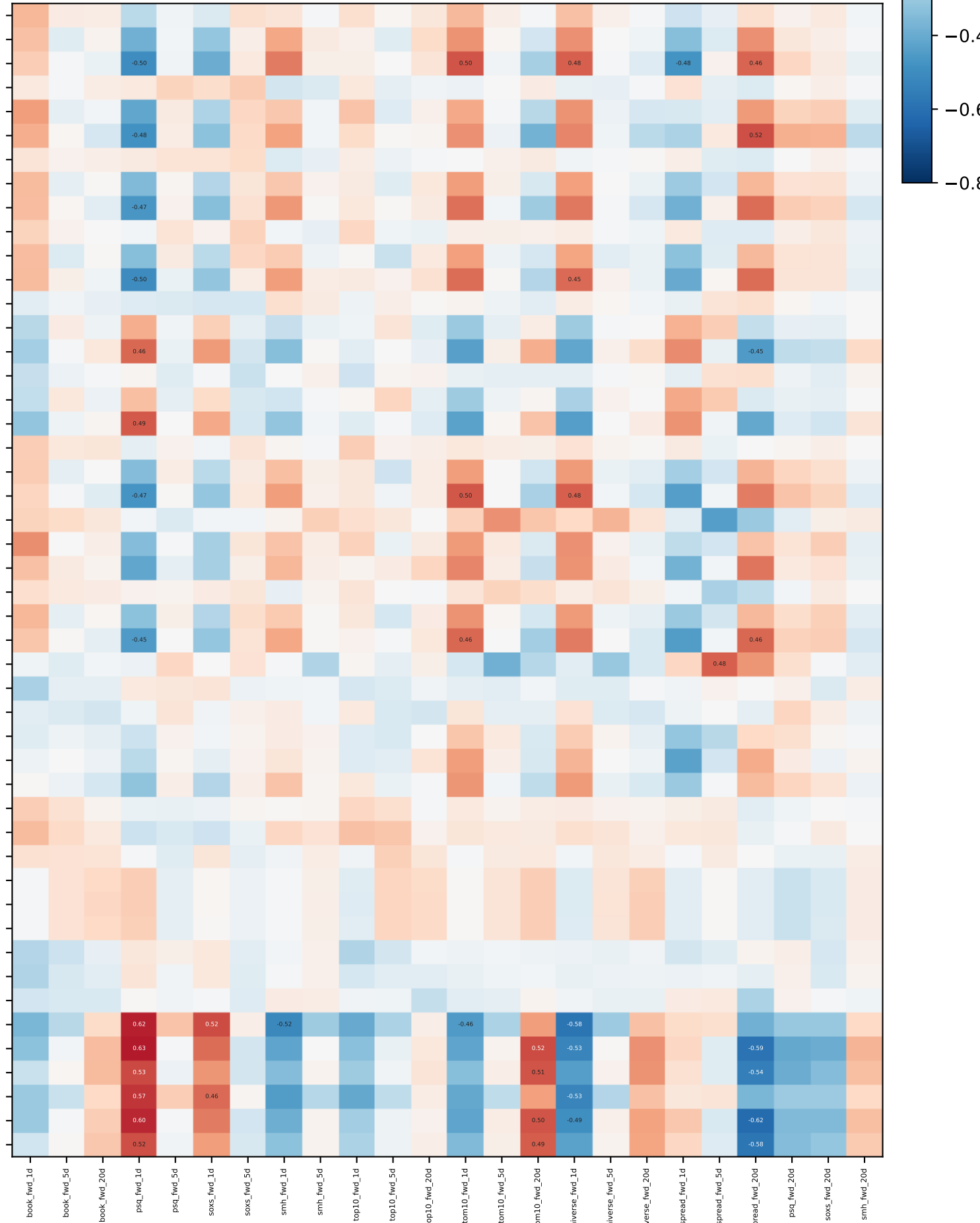
UP > +3.5% (n=68)



DOWN < -2.0% (n=250)



DOWN < -3.5% (n=77)



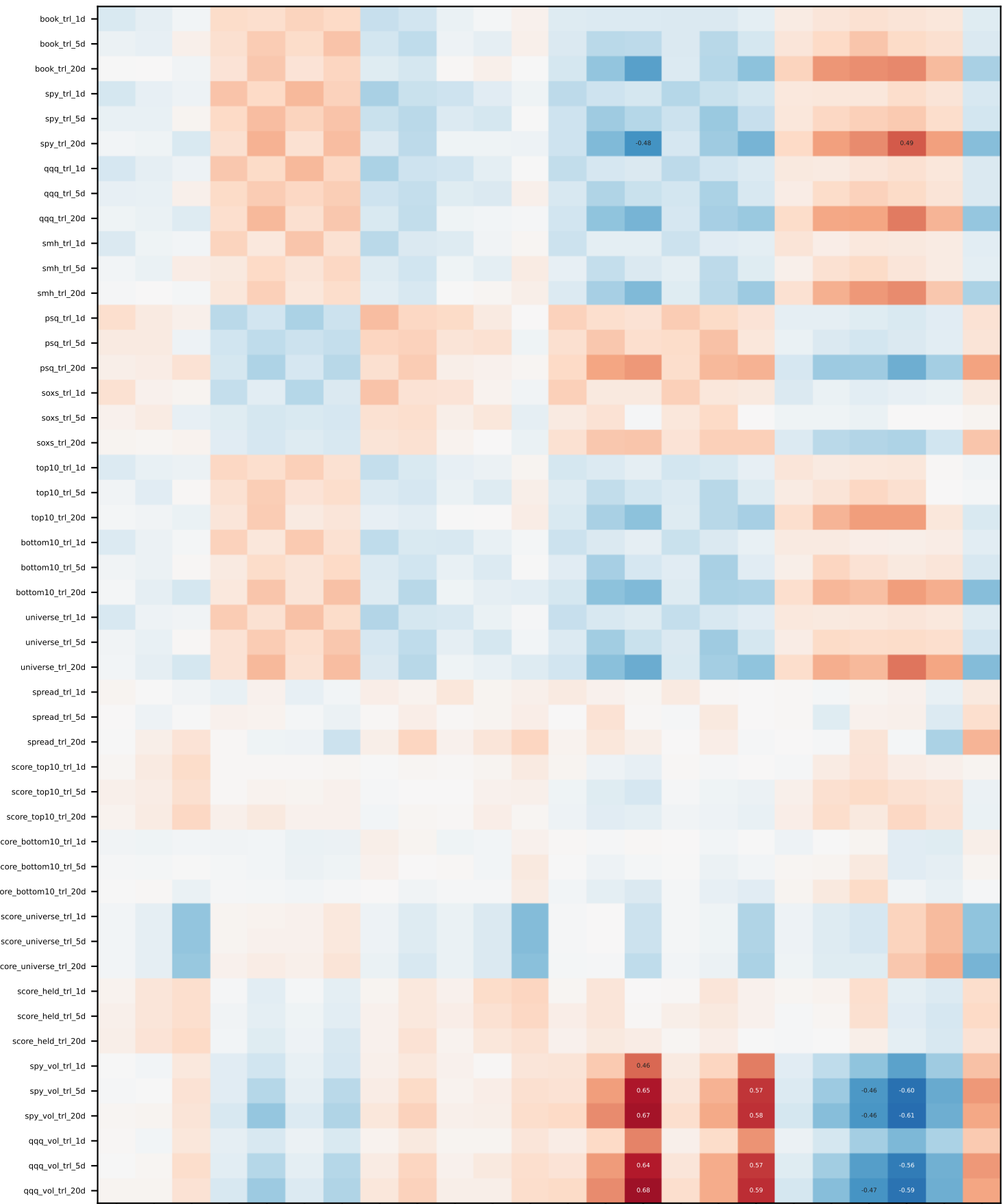
Pearson r



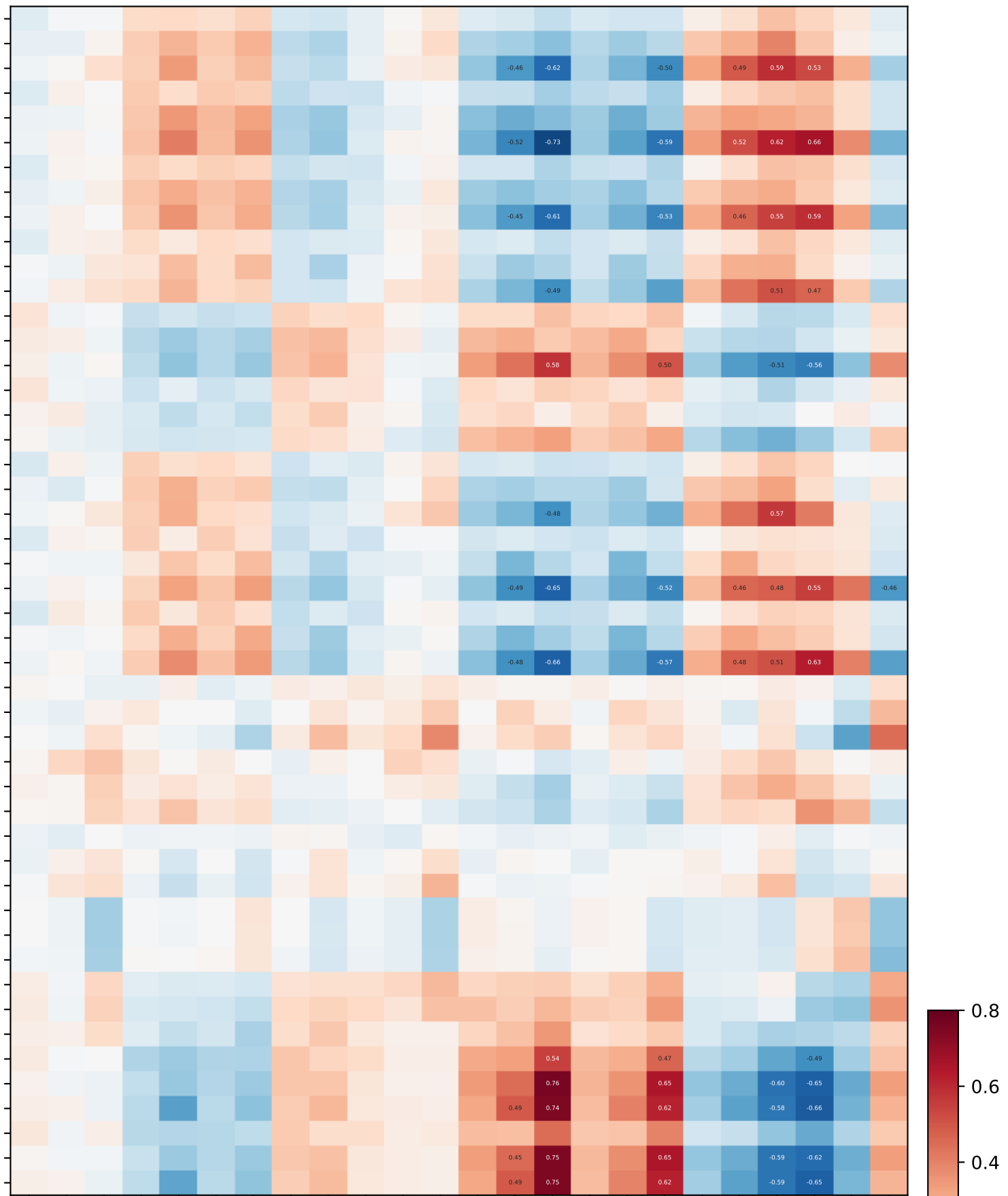




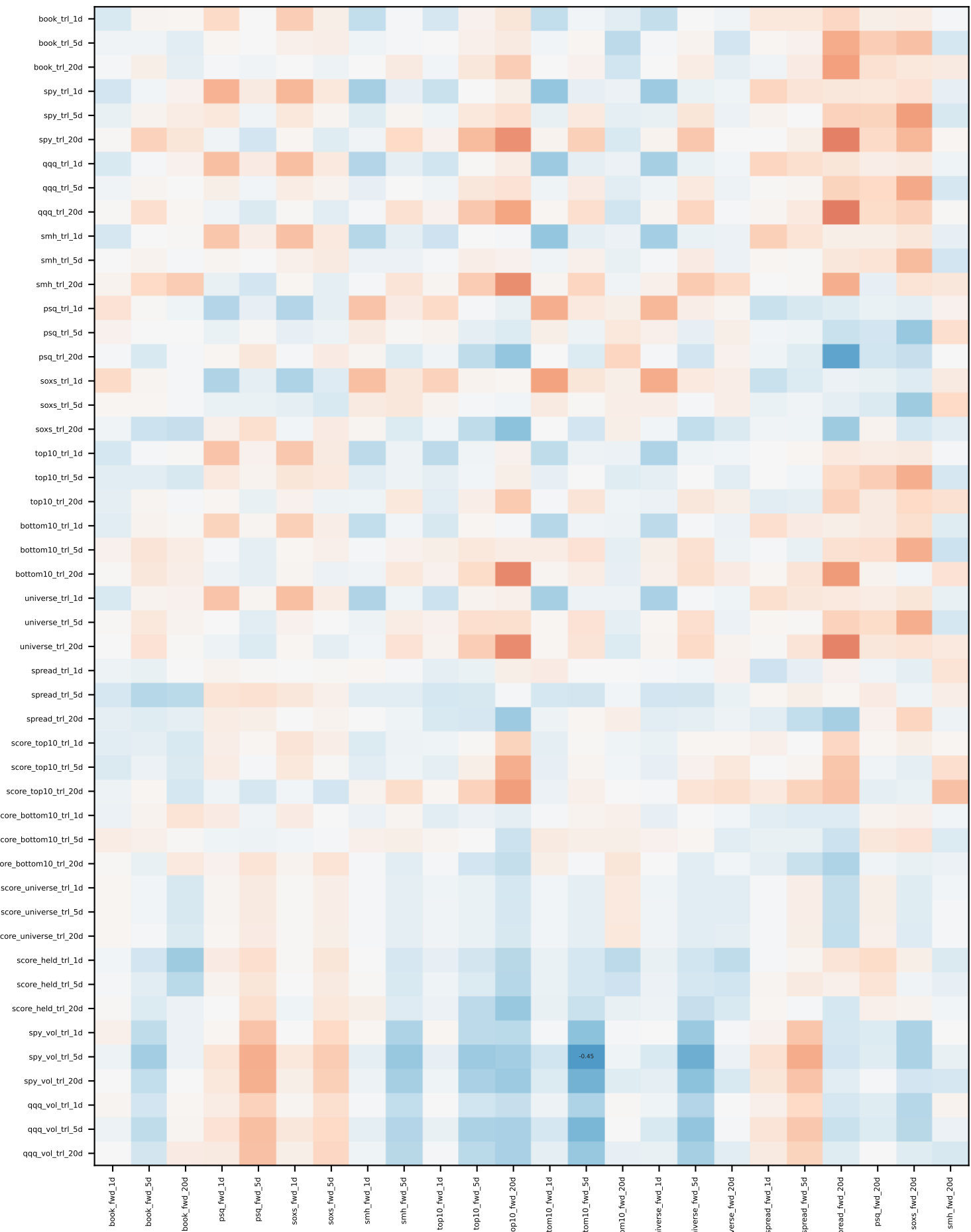
UP > +7.5% (n=655)



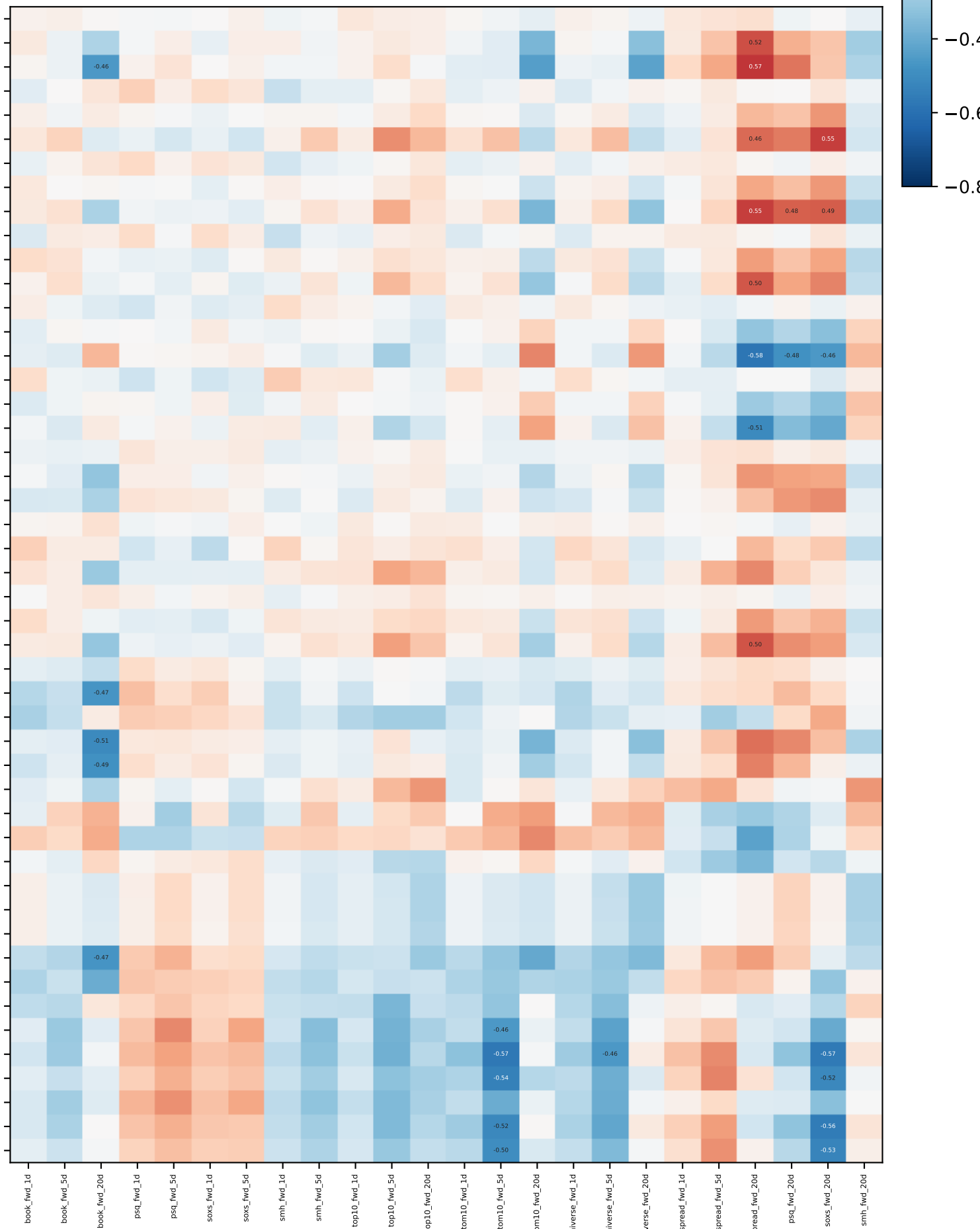
UP > +12.5% (n=268)



DOWN < -7.5% (n=215)



DOWN < -12.5% (n=70)



Pearson r

# Appendix B

Correlation & Gap Analysis (since-COVID, vol-normalized)

*(source: corr\_tails\_by\_horizon\_since\_covid\_volz.pdf)*

D = UP minus DN. Returns/vol abs in %; scores raw.

JUN-15 z shows how far that factor sat above/below its recent norm at that close; compare its sign/size to the UP vs DN segment z to read the regime.

Biggest UP-DN gaps, sorted by |z-gap| within each horizon

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z (value-recent mean)/recent std, L=63d | JUN-15 = 2026-06-15 close

Sorted by absolute normalized gap. Dz colored; (abs gap | zUP/zDN) grey. JUN15 = factor z at 2026-06-15.

FWD 1D (thr +/-2.0%)

| factor             | Dz15z    | (absGap zU/zD)     |
|--------------------|----------|--------------------|
| 1.sc.held_20d      | +0.20.00 | +0.004 +0.28/+0.08 |
| 2.spy_vol_5d       | -0.11.46 | -0.6 +0.03/+0.21   |
| 3.sc.universe_20d  | +0.10.51 | +0.000 -0.57/-0.74 |
| 4.spy_1d           | -0.11.65 | -0.2 -0.15/+0.01   |
| 5.psq_1d           | +0.11.98 | +0.2 +0.17/+0.01   |
| 6.qqq_vol_5d       | -0.12.20 | -0.9 +0.08/+0.23   |
| 7.qqq_1d           | -0.12.02 | -0.2 -0.16/-0.02   |
| 8.soxs_1d          | +0.11.40 | +1.0 +0.16/+0.02   |
| 9.smh_1d           | -0.11.30 | -0.3 -0.16/-0.02   |
| 10.soxs_5d         | +0.11.07 | +2.1 +0.17/+0.04   |
| 11.sc.bottom10_20d | +0.10.38 | +0.001 +0.09/-0.02 |
| 12.sc.universe_5d  | +0.10.80 | +0.000 -0.40/-0.50 |
| 13.sc.top10_20d    | +0.10.23 | +0.001 +0.07/-0.03 |
| 14.smh_5d          | -0.10.84 | -0.5 -0.17/-0.08   |
| 15.book_5d         | -0.10.49 | -0.4 -0.22/-0.12   |
| 16.spread_1d       | -0.01.20 | -0.2 -0.14/-0.05   |
| 17.book_20d        | +0.00.07 | +0.9 +0.01/-0.08   |
| 18.psq_5d          | +0.00.82 | +0.2 +0.22/+0.14   |
| 19.universe_20d    | +0.00.28 | +0.4 -0.14/-0.23   |
| 20.spy_20d         | +0.00.18 | +0.1 -0.25/-0.33   |
| 21.top10_5d        | -0.00.46 | -0.3 -0.16/+0.07   |
| 22.smh_20d         | +0.00.21 | +0.3 -0.10/+0.18   |
| 23.qqq_5d          | -0.00.78 | -0.2 -0.22/-0.14   |
| 24.sc.bottom10_5d  | +0.01.28 | +0.002 -0.02/-0.09 |
| 25.universe_1d     | -0.01.43 | -0.1 -0.16/-0.09   |
| 26.sc.held_1d      | +0.00.30 | +0.002 +0.12/+0.05 |
| 27.bottom10_20d    | +0.00.22 | +0.5 -0.13/-0.19   |
| 28.spread_5d       | -0.00.70 | -0.2 +0.00/+0.06   |
| 29.soxs_20d        | -0.00.71 | -1.1 +0.10/+0.15   |
| 30.sc.held_5d      | +0.01.09 | +0.001 +0.14/+0.09 |
| 31.qqq_vol_1d      | +0.02.22 | +0.5 +0.09/+0.05   |
| 32.spread_20d      | -0.00.50 | +0.2 +0.01/+0.05   |
| 33.spy_5d          | -0.00.61 | -0.1 -0.22/-0.18   |
| 34.top10_1d        | -0.01.72 | -0.1 -0.15/-0.12   |
| 35.sc.universe_1d  | -0.01.26 | +0.000 -0.41/-0.38 |
| 36.qqq_20d         | +0.00.15 | +0.1 -0.23/-0.27   |
| 37.psq_20d         | -0.00.04 | -0.2 +0.24/+0.28   |
| 38.qqq_vol_20d     | -0.02.52 | +1.2 +0.17/+0.20   |
| 39.universe_5d     | -0.00.10 | -0.2 -0.23/-0.20   |
| 40.sc.bottom10_1d  | -0.01.35 | +0.001 -0.03/-0.00 |
| 41.book_1d         | -0.01.26 | -0.2 -0.11/-0.09   |
| 42.spy_vol_20d     | -0.00.51 | +1.5 +0.07/+0.08   |
| 43.bottom10_5d     | -0.00.29 | -0.1 -0.22/-0.21   |
| 44.sc.top10_1d     | -0.00.41 | -0.000 +0.03/+0.04 |
| 45.sc.top10_5d     | -0.01.03 | +0.001 +0.01/+0.02 |
| 46.top10_20d       | +0.00.24 | +0.7 -0.04/-0.05   |
| 47.spy_vol_1d      | -0.01.62 | -0.1 +0.07/+0.07   |
| 48.bottom10_1d     | -0.00.56 | +0.1 -0.06/-0.06   |

FWD 5D (thr +/-4.5%)

| factor             | Dz15z    | (absGap zU/zD)     |
|--------------------|----------|--------------------|
| 1.sc.top10_20d     | +0.40.23 | +0.002 +0.10/-0.30 |
| 2.book_20d         | +0.34.07 | +1.8 +0.13/-0.21   |
| 3.sc.held_20d      | +0.33.00 | +0.006 +0.32/-0.01 |
| 4.universe_20d     | +0.26.28 | +0.5 -0.01/-0.27   |
| 5.sc.universe_20d  | +0.20.51 | +0.000 -0.58/-0.82 |
| 6.sc.universe_5d   | +0.20.80 | +0.000 -0.39/-0.62 |
| 7.sc.held_1d       | +0.20.30 | +0.012 +0.18/-0.04 |
| 8.spy_vol_5d       | -0.21.46 | +1.5 +0.07/+0.29   |
| 9.sc.held_5d       | +0.21.09 | +0.009 +0.24/+0.03 |
| 10.top10_20d       | +0.20.24 | +1.5 +0.10/-0.10   |
| 11.sc.top10_5d     | +0.13.03 | +0.003 +0.03/-0.15 |
| 12.smh_20d         | +0.15.21 | +0.7 -0.09/-0.24   |
| 13.sc.bottom10_20d | +0.10.38 | +0.001 +0.15/-0.00 |
| 14.bottom10_20d    | +0.10.22 | +0.3 -0.10/-0.24   |
| 15.spread_20d      | +0.10.50 | +1.2 +0.15/+0.01   |
| 16.sc.top10_1d     | +0.10.41 | +0.003 +0.02/-0.12 |
| 17.spy_20d         | +0.10.18 | -0.0 -0.15/-0.29   |
| 18.book_1d         | -0.11.26 | -0.2 -0.10/+0.03   |
| 19.qqq_vol_5d      | -0.12.20 | +1.1 +0.13/+0.26   |
| 20.qqq_20d         | +0.10.15 | +0.1 -0.13/-0.26   |
| 21.spy_vol_1d      | -0.12.62 | +0.7 +0.06/+0.18   |
| 22.sc.bottom10_1d  | -0.12.35 | -0.002 +0.10/+0.22 |
| 23.psq_20d         | -0.10.04 | -0.1 +0.15/+0.26   |
| 24.spy_1d          | -0.11.65 | -0.1 -0.07/+0.04   |
| 25.sc.universe_1d  | +0.11.26 | +0.000 -0.28/-0.39 |
| 26.psq_1d          | +0.01.98 | +0.1 +0.05/-0.04   |
| 27.qqq_1d          | -0.02.02 | -0.1 -0.05/+0.03   |
| 28.spread_5d       | +0.00.70 | +0.3 +0.06/-0.01   |
| 29.spy_5d          | -0.00.61 | -0.3 -0.17/-0.10   |
| 30.top10_1d        | -0.01.72 | -0.1 -0.04/+0.03   |
| 31.book_5d         | -0.00.49 | -0.2 -0.19/-0.12   |
| 32.soxs_5d         | +0.01.07 | +1.2 +0.12/+0.07   |
| 33.soxs_1d         | +0.01.40 | +0.3 -0.02/-0.07   |
| 34.spread_1d       | -0.01.20 | -0.1 +0.03/+0.08   |
| 35.spy_vol_20d     | +0.00.51 | +3.9 +0.13/+0.08   |
| 36.smh_1d          | -0.01.30 | -0.1 +0.01/+0.06   |
| 37.universe_1d     | -0.01.43 | -0.1 -0.05/-0.00   |
| 38.psq_5d          | +0.00.82 | +0.3 +0.16/+0.12   |
| 39.sc.bottom10_5d  | -0.01.28 | +0.000 +0.11/+0.15 |
| 40.qqq_vol_1d      | -0.02.22 | +0.7 +0.09/+0.13   |
| 41.qqq_5d          | -0.00.78 | -0.3 -0.15/-0.12   |
| 42.bottom10_5d     | +0.00.29 | -0.5 -0.14/-0.17   |
| 43.top10_5d        | -0.00.46 | -0.3 -0.09/-0.06   |
| 44.universe_5d     | +0.00.10 | -0.4 -0.14/-0.16   |
| 45.soxs_20d        | -0.00.71 | -1.0 +0.16/+0.18   |
| 46.qqq_vol_20d     | +0.02.52 | +2.8 +0.20/+0.19   |
| 47.bottom10_1d     | +0.00.56 | +0.0 -0.08/-0.09   |
| 48.smh_5d          | -0.00.84 | -0.2 -0.10/-0.10   |

FWD 20D (thr +/-7.5%)

| factor             | Dz15z    | (absGap zU/zD)     |
|--------------------|----------|--------------------|
| 1.book_20d         | +0.54.07 | +3.5 +0.25/-0.29   |
| 2.sc.top10_5d      | +0.46.03 | +0.008 +0.11/-0.35 |
| 3.sc.top10_20d     | +0.42.23 | +0.001 +0.10/-0.33 |
| 4.sc.held_5d       | +0.31.09 | +0.005 +0.14/-0.17 |
| 5.sc.top10_1d      | +0.30.41 | +0.007 +0.11/-0.18 |
| 6.sc.held_1d       | +0.20.30 | +0.007 +0.14/-0.11 |
| 7.universe_20d     | +0.20.28 | +1.2 +0.12/-0.13   |
| 8.spy_vol_20d      | +0.20.51 | +2.8 +0.11/-0.12   |
| 9.sc.bottom10_20d  | +0.20.38 | +0.002 +0.22/+0.00 |
| 10.top10_20d       | +0.21.24 | +2.6 +0.26/+0.05   |
| 11.sc.held_20d     | +0.20.00 | -0.007 +0.18/-0.02 |
| 12.bottom10_20d    | +0.10.22 | +2.3 +0.03/-0.15   |
| 13.sc.universe_20d | -0.10.51 | +0.000 -0.61/-0.44 |
| 14.smh_20d         | +0.10.21 | +1.4 -0.04/-0.20   |
| 15.spy_vol_5d      | +0.11.46 | +2.8 +0.00/-0.14   |
| 16.sc.universe_5d  | -0.10.80 | +0.000 -0.46/-0.33 |
| 17.spread_5d       | +0.10.70 | +0.7 +0.14/+0.01   |
| 18.book_5d         | +0.10.49 | +0.5 +0.13/+0.00   |
| 19.psq_20d         | -0.10.04 | -0.8 +0.04/+0.14   |
| 20.sc.bottom10_5d  | +0.11.28 | +0.002 +0.13/+0.03 |
| 21.qqq_20d         | +0.00.15 | +0.7 -0.03/-0.12   |
| 22.soxs_5d         | +0.01.07 | +0.5 +0.02/-0.07   |
| 23.bottom10_5d     | -0.00.29 | -0.4 -0.04/+0.03   |
| 24.sc.bottom10_1d  | -0.01.35 | -0.001 +0.07/+0.14 |
| 25.spy_5d          | -0.00.61 | -0.1 -0.01/+0.06   |
| 26.bottom10_1d     | +0.00.56 | +0.1 +0.01/-0.06   |
| 27.spy_vol_1d      | +0.01.62 | +2.2 +0.03/-0.03   |
| 28.spread_1d       | -0.01.20 | -0.1 +0.07/+0.12   |
| 29.universe_1d     | +0.01.43 | +0.0 +0.05/-0.00   |
| 30.qqq_5d          | -0.00.78 | -0.1 -0.02/+0.02   |
| 31.psq_5d          | +0.00.82 | +0.1 +0.02/-0.02   |
| 32.sc.universe_1d  | -0.01.26 | +0.000 -0.36/-0.32 |
| 33.smh_5d          | -0.00.84 | +0.0 +0.01/+0.05   |
| 34.qqq_vol_20d     | +0.02.52 | +0.8 +0.16/+0.12   |
| 35.smh_1d          | +0.01.30 | +0.1 +0.05/+0.02   |
| 36.soxs_20d        | -0.00.71 | -2.9 +0.10/+0.13   |
| 37.qqq_vol_1d      | -0.02.22 | +0.1 +0.01/+0.03   |
| 38.book_1d         | +0.01.26 | +0.1 +0.09/+0.06   |
| 39.universe_5d     | -0.00.10 | -0.3 +0.03/+0.05   |
| 40.spy_20d         | +0.00.18 | +0.5 -0.02/-0.04   |
| 41.top10_5d        | +0.00.46 | +0.2 +0.14/+0.12   |
| 42.soxs_1d         | -0.01.40 | -0.1 -0.04/-0.02   |
| 43.spy_1d          | +0.01.65 | +0.0 +0.02/+0.00   |
| 44.psq_1d          | -0.01.98 | -0.0 -0.02/+0.00   |
| 45.qqq_1d          | +0.02.02 | +0.0 +0.02/+0.00   |
| 46.top10_1d        | -0.01.72 | -0.0 +0.09/+0.11   |
| 47.spread_20d      | -0.00.50 | +0.4 +0.07/+0.08   |
| 48.qqq_vol_5d      | -0.02.20 | +0.6 -0.01/-0.01   |

# Normalized trailing(z) -> raw forward correlation call-outs (|r| >= 0.5)

SINCE COVID (2020-04-01 to 2026-06-18, n=1562) | z = rolling z (value-recent mean)/recent std, L=63d | JUN-15 = 2026-06-15 close

z=mean trailing z in panel; J15=factor z at 2026-06-15; avg=mean raw forward return. Small-n (n<~40) noisy.

## Forward 1-Day portfolio return (filtered by book\_fwd\_1d)

|                       |                |         |           |                    |            |
|-----------------------|----------------|---------|-----------|--------------------|------------|
| UP > +2.0% (n=230)    |                |         |           |                    |            |
| (none  r >=0.5)       |                |         |           |                    |            |
| UP > +3.5% (n=52)     |                |         |           |                    |            |
| MODERATE (0.50-0.70): |                |         |           |                    |            |
| +0.55                 | universe_5d    | z=-0.26 | J15=-0.10 | -> spread_fwd_1d   | avg=+0.6%  |
| -0.53                 | soxs_5d        | z=+0.12 | J15=-1.07 | -> spread_fwd_1d   | avg=+0.6%  |
| +0.52                 | book_5d        | z=-0.10 | J15=+0.49 | -> spread_fwd_1d   | avg=+0.6%  |
| +0.52                 | top10_5d       | z=-0.07 | J15=+0.46 | -> spread_fwd_1d   | avg=+0.6%  |
| +0.51                 | book_1d        | z=-0.09 | J15=+1.26 | -> soxs_fwd_5d     | avg=-12.6% |
| +0.51                 | sc.bottom10_5d | z=-0.26 | J15=+1.28 | -> bottom10_fwd_5d | avg=+2.6%  |
| +0.50                 | spy_vol_20d    | z=+0.18 | J15=+0.51 | -> smh_fwd_1d      | avg=+3.6%  |
| +0.50                 | qqq_5d         | z=-0.25 | J15=+0.78 | -> spread_fwd_1d   | avg=+0.6%  |
| -0.50                 | psq_5d         | z=+0.27 | J15=-0.82 | -> spread_fwd_1d   | avg=+0.6%  |
| DOWN < -2.0% (n=183)  |                |         |           |                    |            |
| (none  r >=0.5)       |                |         |           |                    |            |
| DOWN < -3.5% (n=56)   |                |         |           |                    |            |
| MODERATE (0.50-0.70): |                |         |           |                    |            |
| -0.50                 | universe_20d   | z=-0.37 | J15=+0.28 | -> spread_fwd_1d   | avg=-0.3%  |

## Forward 5-Day portfolio return (filtered by book\_fwd\_5d)

|                       |             |         |           |                   |           |
|-----------------------|-------------|---------|-----------|-------------------|-----------|
| UP > +4.5% (n=269)    |             |         |           |                   |           |
| (none  r >=0.5)       |             |         |           |                   |           |
| UP > +6.5% (n=104)    |             |         |           |                   |           |
| MODERATE (0.50-0.70): |             |         |           |                   |           |
| +0.51                 | sc.top10_5d | z=+0.10 | J15=-1.03 | -> spread_fwd_20d | avg=+1.8% |
| DOWN < -4.5% (n=162)  |             |         |           |                   |           |
| (none  r >=0.5)       |             |         |           |                   |           |
| DOWN < -6.5% (n=72)   |             |         |           |                   |           |
| MODERATE (0.50-0.70): |             |         |           |                   |           |
| +0.58                 | sc.held_5d  | z=+0.13 | J15=-1.09 | -> spread_fwd_20d | avg=-2.9% |

## Forward 20-Day portfolio return (filtered by book\_fwd\_20d)

|                       |                 |         |           |                     |            |
|-----------------------|-----------------|---------|-----------|---------------------|------------|
| UP > +7.5% (n=422)    |                 |         |           |                     |            |
| (none  r >=0.5)       |                 |         |           |                     |            |
| UP > +12.5% (n=209)   |                 |         |           |                     |            |
| (none  r >=0.5)       |                 |         |           |                     |            |
| DOWN < -7.5% (n=161)  |                 |         |           |                     |            |
| MODERATE (0.50-0.70): |                 |         |           |                     |            |
| +0.59                 | spy_20d         | z=-0.04 | J15=-0.18 | -> spread_fwd_20d   | avg=-2.5%  |
| +0.54                 | universe_20d    | z=-0.13 | J15=+0.28 | -> spread_fwd_20d   | avg=-2.5%  |
| DOWN < -12.5% (n=42)  |                 |         |           |                     |            |
| STRONG ( r >=0.70):   |                 |         |           |                     |            |
| +0.79                 | spy_20d         | z=-0.33 | J15=-0.18 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.71                 | psq_20d         | z=+0.45 | J15=+0.04 | -> spread_fwd_20d   | avg=-2.4%  |
| MODERATE (0.50-0.70): |                 |         |           |                     |            |
| +0.69                 | qqq_20d         | z=-0.43 | J15=-0.15 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.67                 | spy_20d         | z=-0.33 | J15=-0.18 | -> bottom10_fwd_20d | avg=-12.2% |
| +0.66                 | universe_20d    | z=-0.42 | J15=+0.28 | -> spread_fwd_20d   | avg=-2.4%  |
| +0.66                 | psq_5d          | z=-0.05 | J15=-0.82 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.66                 | qqq_5d          | z=+0.05 | J15=+0.78 | -> bottom10_fwd_20d | avg=-12.2% |
| +0.66                 | book_5d         | z=-0.12 | J15=+0.49 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.66                 | sc.bottom10_20d | z=-0.07 | J15=-0.38 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.64                 | sc.top10_20d    | z=-0.25 | J15=+0.23 | -> book_fwd_20d     | avg=-16.4% |



|       |                 |         |           |                     |            |
|-------|-----------------|---------|-----------|---------------------|------------|
| +0.64 | bottom10_20d    | z=-0.47 | J15=-0.22 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.64 | soxs_20d        | z=+0.53 | J15=-0.71 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.63 | sc.bottom10_20d | z=-0.07 | J15=-0.38 | -> psq_fwd_20d      | avg=+9.9%  |
| +0.62 | spy_20d         | z=-0.33 | J15=-0.18 | -> spread_fwd_5d    | avg=+0.5%  |
| +0.62 | sc.bottom10_20d | z=-0.07 | J15=-0.38 | -> bottom10_fwd_20d | avg=-12.2% |
| +0.61 | book_20d        | z=-0.68 | J15=-0.07 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.61 | spy_5d          | z=+0.05 | J15=+0.61 | -> bottom10_fwd_20d | avg=-12.2% |
| +0.61 | qqq_5d          | z=+0.05 | J15=+0.78 | -> spread_fwd_20d   | avg=-2.4%  |
| +0.61 | smh_20d         | z=-0.67 | J15=+0.21 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.61 | psq_5d          | z=-0.05 | J15=-0.82 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.61 | book_5d         | z=-0.12 | J15=+0.49 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.60 | smh_5d          | z=+0.13 | J15=+0.84 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.60 | spread_5d       | z=+0.21 | J15=+0.70 | -> book_fwd_20d     | avg=-16.4% |
| -0.59 | spy_vol_5d      | z=+0.12 | J15=+1.46 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.59 | sc.top10_5d     | z=-0.50 | J15=-1.03 | -> book_fwd_20d     | avg=-16.4% |
| +0.58 | soxs_5d         | z=-0.11 | J15=-1.07 | -> universe_fwd_20d | avg=-12.2% |
| +0.57 | spy_5d          | z=+0.05 | J15=+0.61 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.57 | smh_5d          | z=+0.13 | J15=+0.84 | -> universe_fwd_20d | avg=-12.2% |
| +0.57 | psq_20d         | z=+0.45 | J15=+0.04 | -> bottom10_fwd_20d | avg=-12.2% |
| +0.57 | soxs_5d         | z=-0.11 | J15=-1.07 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.56 | qqq_20d         | z=-0.43 | J15=-0.15 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.56 | universe_5d     | z=-0.03 | J15=-0.10 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.56 | qqq_vol_5d      | z=+0.02 | J15=+2.20 | -> spread_fwd_20d   | avg=-2.4%  |
| +0.56 | psq_5d          | z=-0.05 | J15=-0.82 | -> universe_fwd_20d | avg=-12.2% |
| +0.55 | bottom10_20d    | z=-0.47 | J15=-0.22 | -> spread_fwd_5d    | avg=+0.5%  |
| -0.55 | qqq_5d          | z=+0.05 | J15=+0.78 | -> universe_fwd_20d | avg=-12.2% |
| +0.55 | sc.bottom10_5d  | z=+0.29 | J15=+1.28 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.55 | top10_5d        | z=+0.27 | J15=+0.46 | -> book_fwd_20d     | avg=-16.4% |
| +0.54 | universe_5d     | z=-0.03 | J15=-0.10 | -> spread_fwd_20d   | avg=-2.4%  |
| -0.54 | sc.bottom10_20d | z=-0.07 | J15=-0.38 | -> soxs_fwd_20d     | avg=+35.2% |
| +0.54 | sc.bottom10_20d | z=-0.07 | J15=-0.38 | -> universe_fwd_20d | avg=-12.2% |
| +0.54 | soxs_20d        | z=+0.53 | J15=-0.71 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.54 | smh_20d         | z=-0.67 | J15=+0.21 | -> bottom10_fwd_20d | avg=-12.2% |
| +0.53 | smh_5d          | z=+0.13 | J15=+0.84 | -> spread_fwd_20d   | avg=-2.4%  |
| +0.53 | sc.held_20d     | z=+0.02 | J15=-0.00 | -> smh_fwd_20d      | avg=-12.6% |
| +0.53 | spy_20d         | z=-0.33 | J15=-0.18 | -> psq_fwd_20d      | avg=+9.9%  |
| +0.53 | book_20d        | z=-0.68 | J15=-0.07 | -> spread_fwd_5d    | avg=+0.5%  |
| -0.53 | bottom10_20d    | z=-0.47 | J15=-0.22 | -> book_fwd_20d     | avg=-16.4% |
| -0.53 | sc.bottom10_5d  | z=+0.29 | J15=+1.28 | -> spread_fwd_20d   | avg=-2.4%  |
| +0.52 | bottom10_20d    | z=-0.47 | J15=-0.22 | -> top10_fwd_20d    | avg=-14.6% |
| -0.52 | psq_20d         | z=+0.45 | J15=+0.04 | -> spread_fwd_5d    | avg=+0.5%  |
| +0.52 | soxs_5d         | z=-0.11 | J15=-1.07 | -> smh_fwd_20d      | avg=-12.6% |
| -0.52 | sc.bottom10_5d  | z=+0.29 | J15=+1.28 | -> psq_fwd_20d      | avg=+9.9%  |
| -0.52 | universe_20d    | z=-0.42 | J15=+0.28 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.51 | spy_20d         | z=-0.33 | J15=-0.18 | -> universe_fwd_20d | avg=-12.2% |
| +0.51 | universe_20d    | z=-0.42 | J15=+0.28 | -> spread_fwd_5d    | avg=+0.5%  |
| -0.51 | spy_5d          | z=+0.05 | J15=+0.61 | -> universe_fwd_20d | avg=-12.2% |
| +0.51 | sc.bottom10_5d  | z=+0.29 | J15=+1.28 | -> universe_fwd_20d | avg=-12.2% |
| +0.50 | qqq_20d         | z=-0.43 | J15=-0.15 | -> spread_fwd_5d    | avg=+0.5%  |
| -0.50 | top10_5d        | z=+0.27 | J15=+0.46 | -> bottom10_fwd_20d | avg=-12.2% |
| -0.50 | top10_5d        | z=+0.27 | J15=+0.46 | -> universe_fwd_20d | avg=-12.2% |
| +0.50 | book_20d        | z=-0.68 | J15=-0.07 | -> top10_fwd_20d    | avg=-14.6% |



